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(54) PLASMA PROCESSING DEVICE

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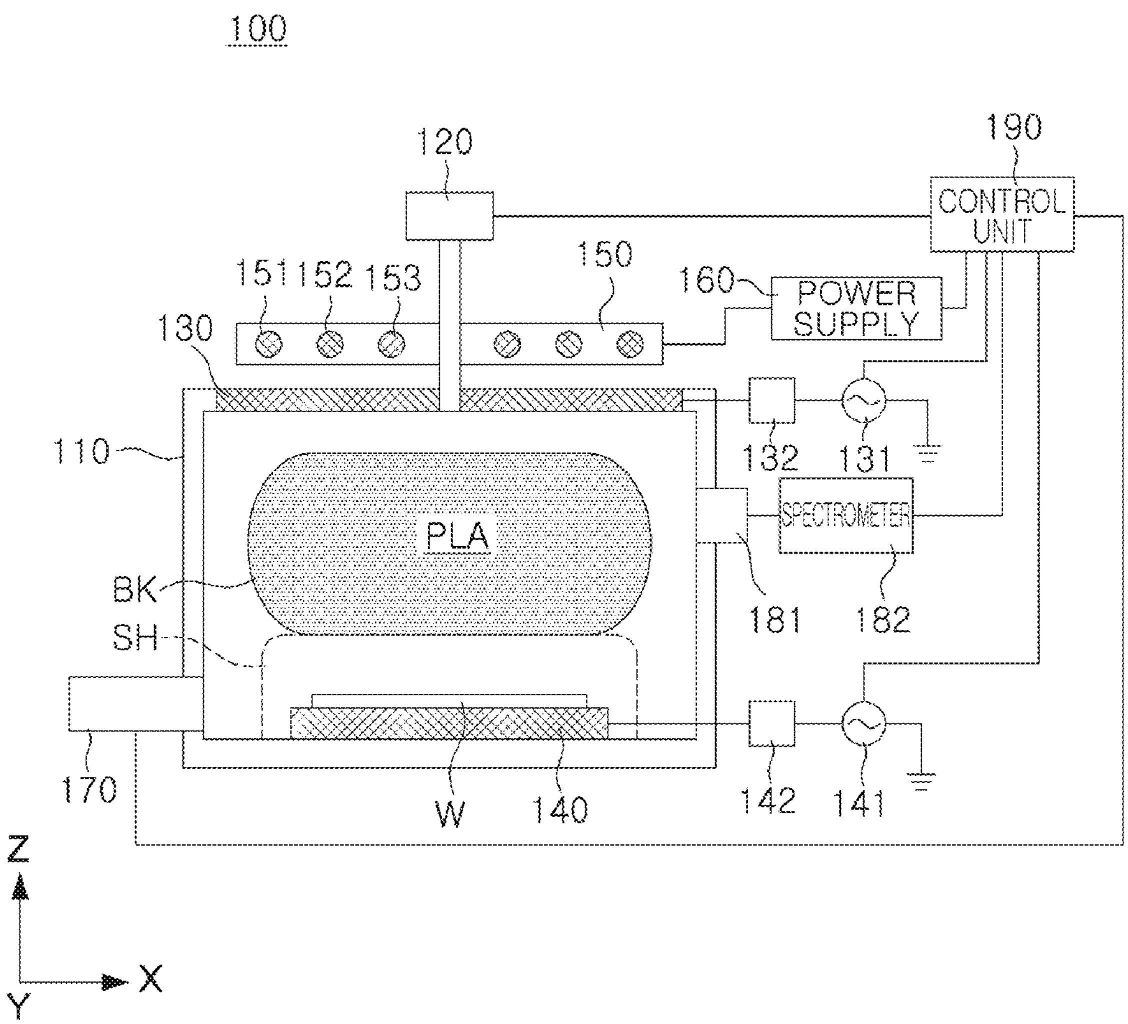
(57) ABSTRACT

A plasma processing device includes an electrostatic chuck supporting a substrate in a chamber space; an upper electrode located in an upper portion of a chamber body; a magnetic field control device including at least one coil located above the upper electrode and forming a magnetic field in the chamber space; and a control unit controlling at least one current flowing through the at least one coil such that a magnetic flux density at a target position on a boundary of the sheath region has a value in which an electronic rotation period at the target position due to the magnetic field matches an electronic oscillation period determined by at least one of a first high-frequency power source and a second high-frequency power source.



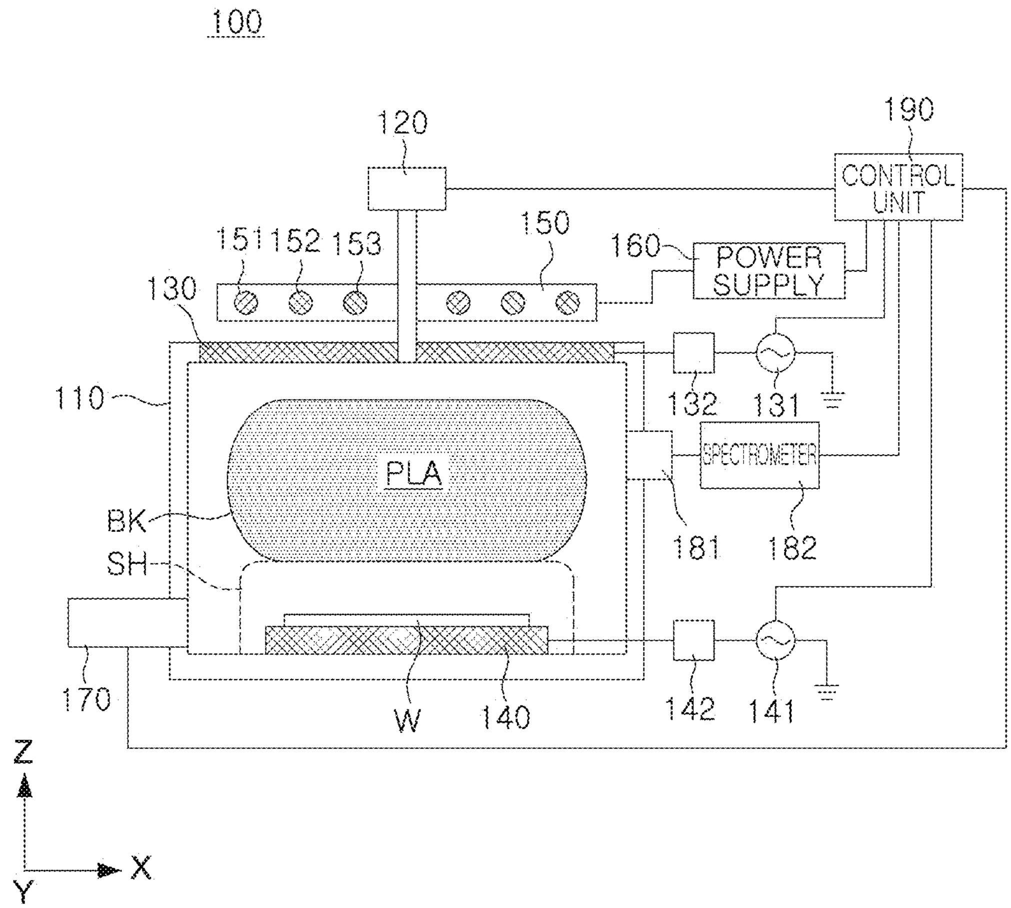

100
120
190
CONTROL UNIT
151
152
153
150
160
POWER SUPPLY
130
110
132
131
PLA
SPECTROMETER
BK
SH
181
182
170
W
140
142
141
Z
X
Y

FIG. 1

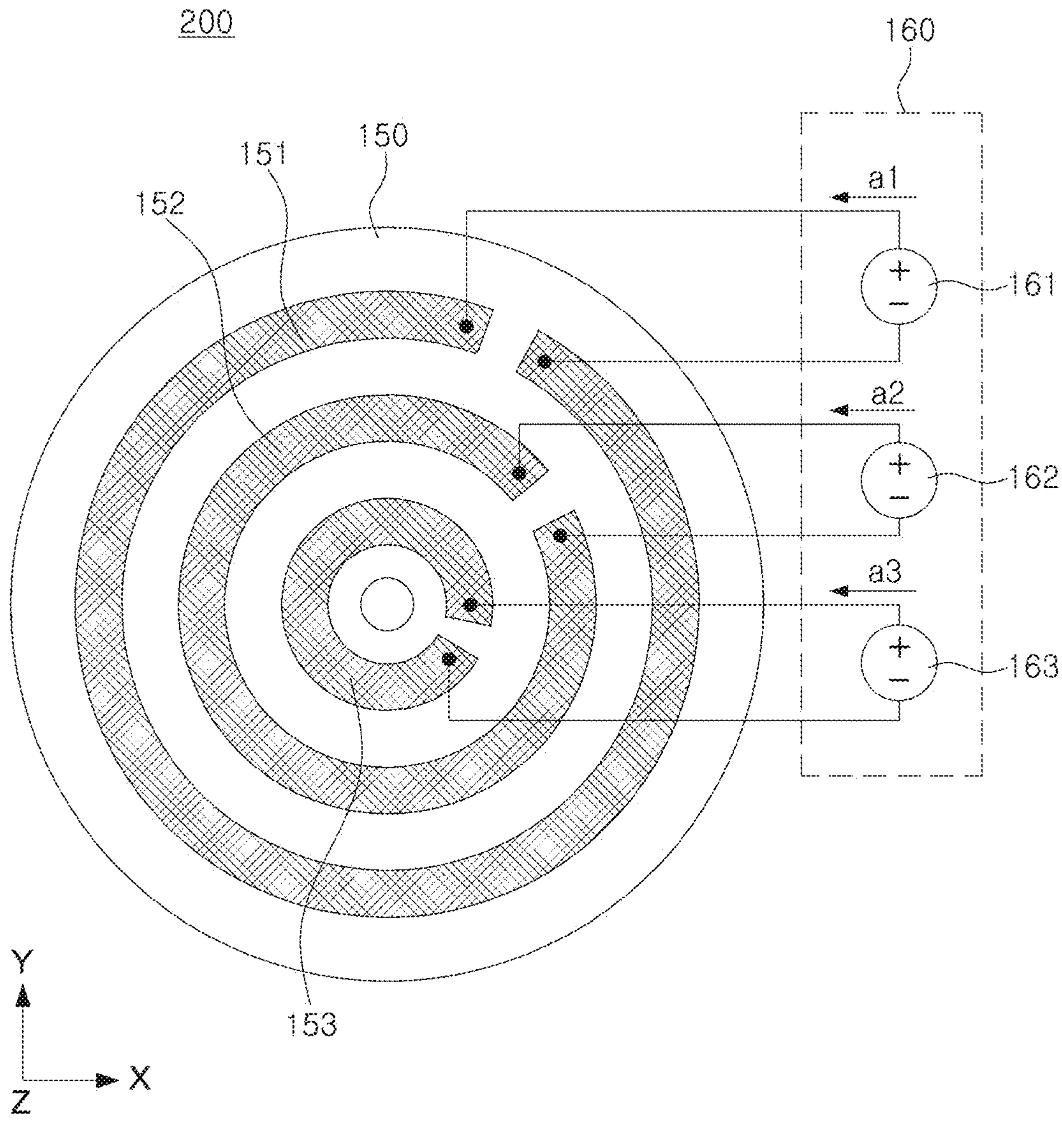


FIG. 2

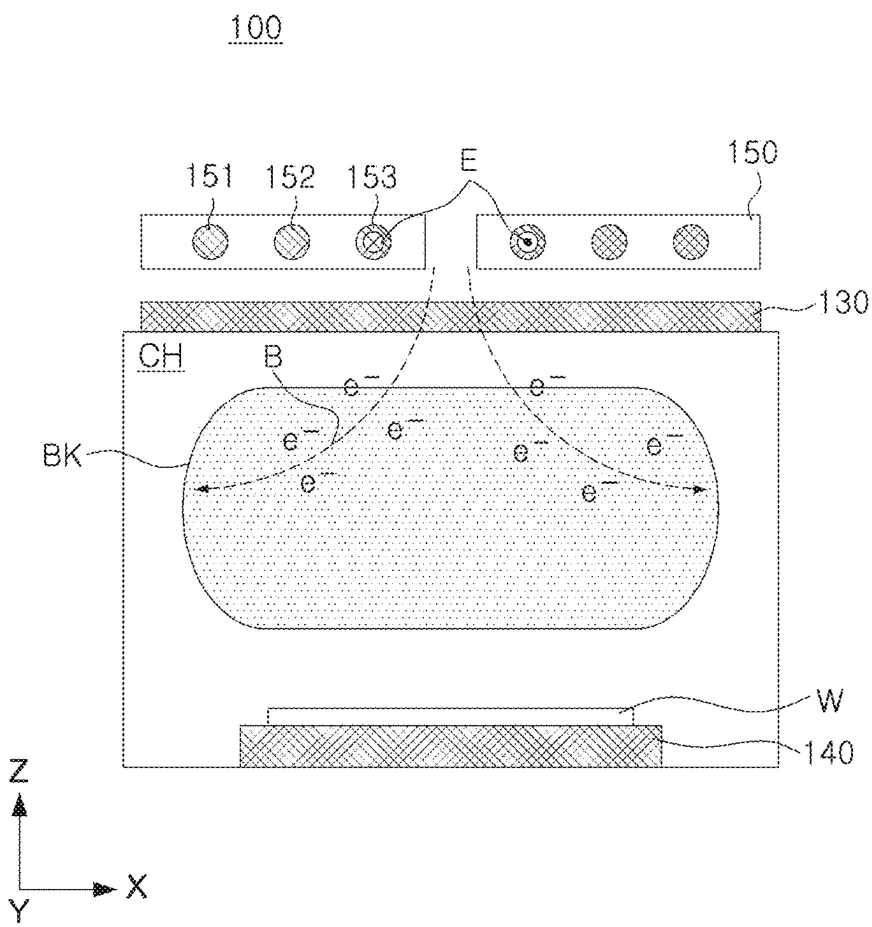


FIG. 3

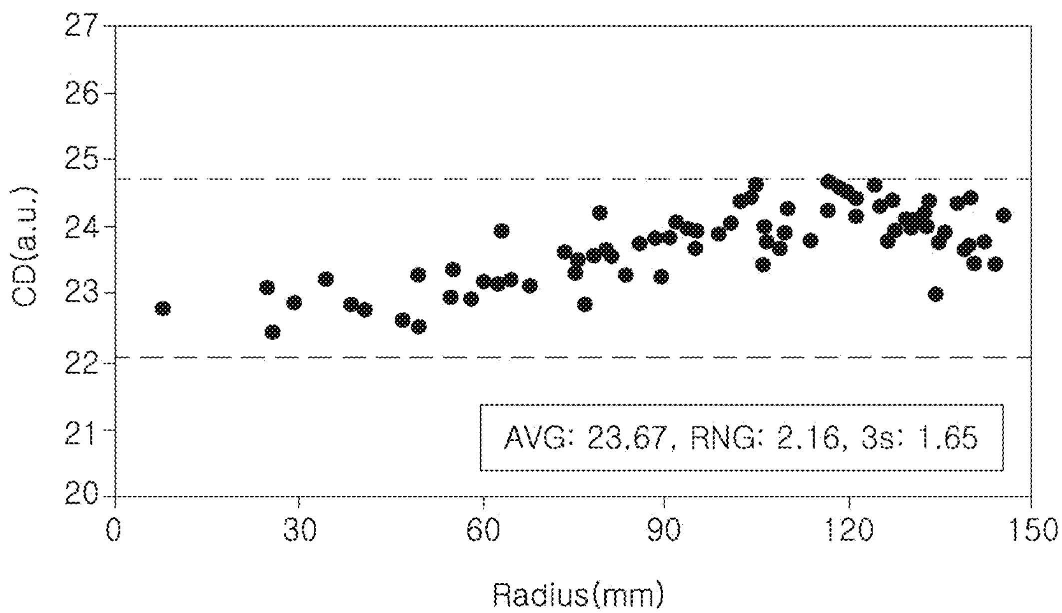

27
26
25
24
23
22
21
20
CD(a.u.)
0
30
60
90
120
150
Radius(mm)
AVG: 23.67, RNG: 2.16, 3s: 1.65

FIG. 4

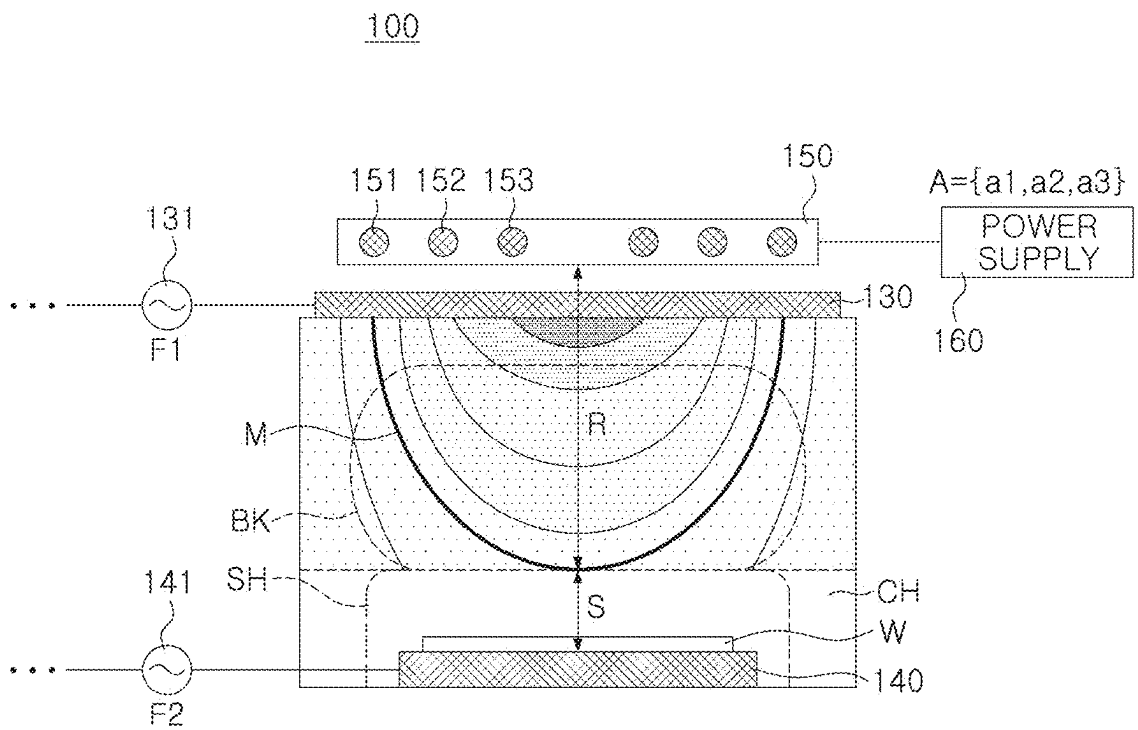

100
151 152 153
150
A={a1,a2,a3}
POWER SUPPLY
131
130
F1
160
M
R
BK
141
SH
S
CH
W
140
F2

FIG. 5

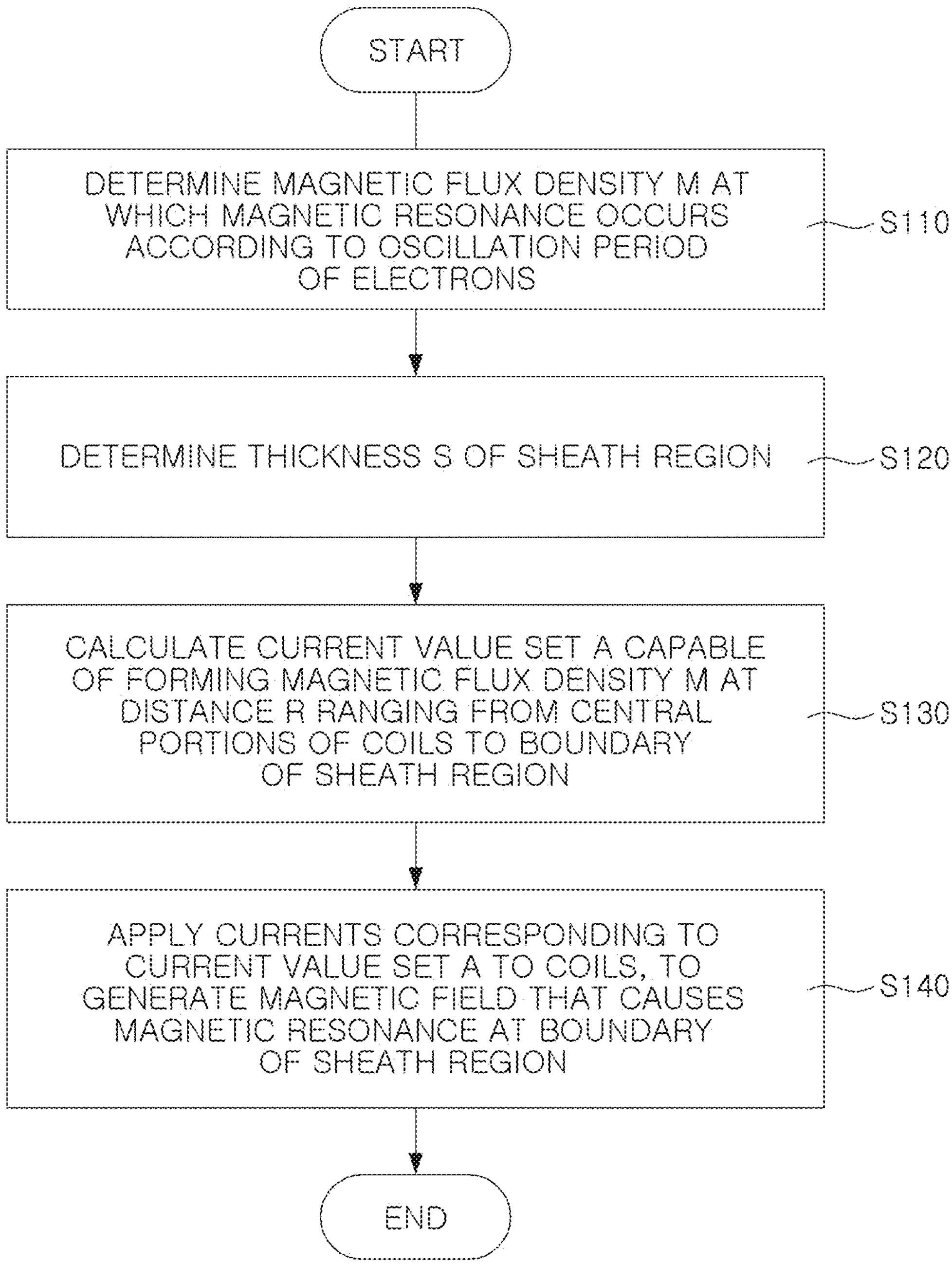

START
DETERMINE MAGNETIC FLUX DENSITY M AT WHICH MAGNETIC RESONANCE OCCURS ACCORDING TO OSCILLATION PERIOD OF ELECTRONS
S110
DETERMINE THICKNESS S OF SHEATH REGION
S120
CALCULATE CURRENT VALUE SET A CAPABLE OF FORMING MAGNETIC FLUX DENSITY M AT DISTANCE R RANGING FROM CENTRAL PORTIONS OF COILS TO BOUNDARY OF SHEATH REGION
S130
APPLY CURRENTS CORRESPONDING TO CURRENT VALUE SET A TO COILS, TO GENERATE MAGNETIC FIELD THAT CAUSES MAGNETIC RESONANCE AT BOUNDARY OF SHEATH REGION
S140
END

FIG. 6

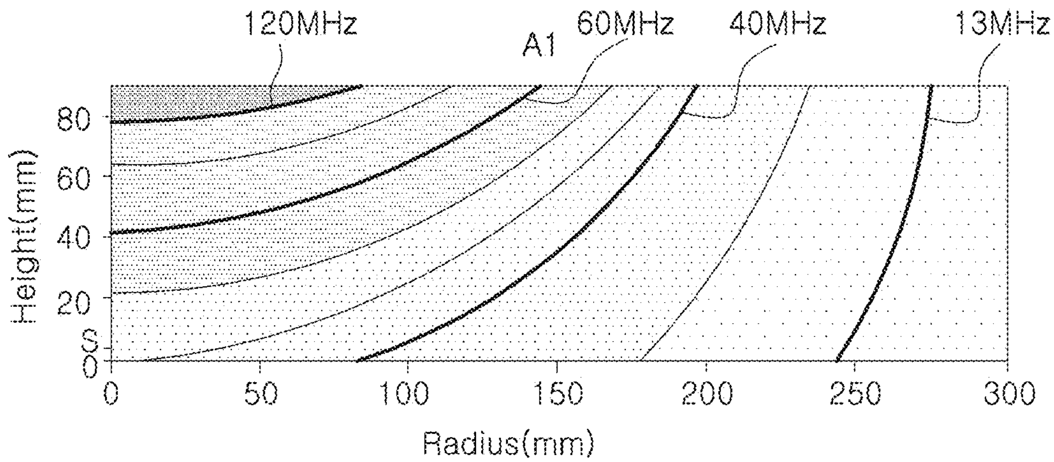


FIG. 7A

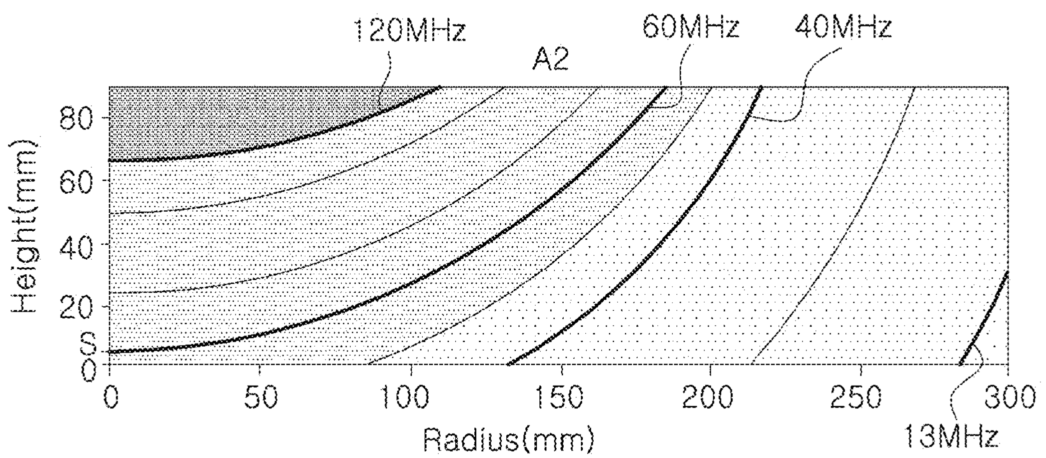


FIG. 7B

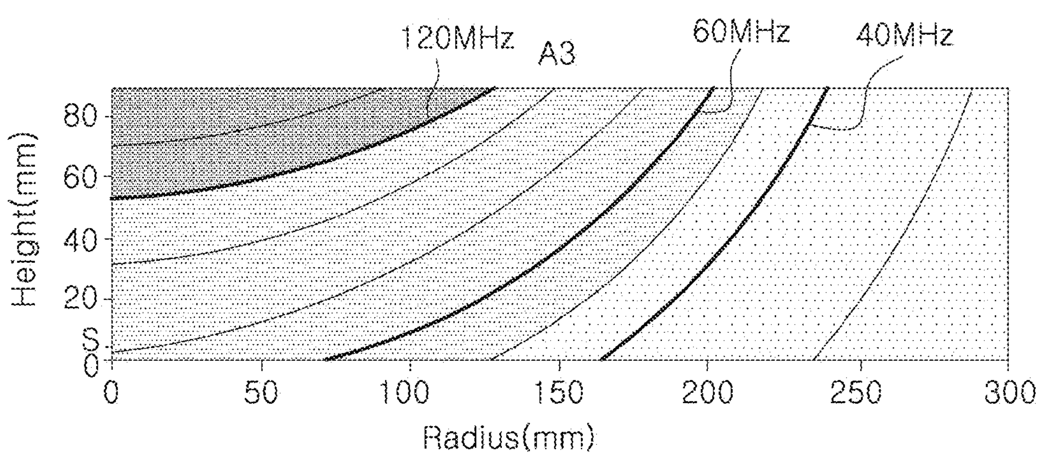


FIG. 7C

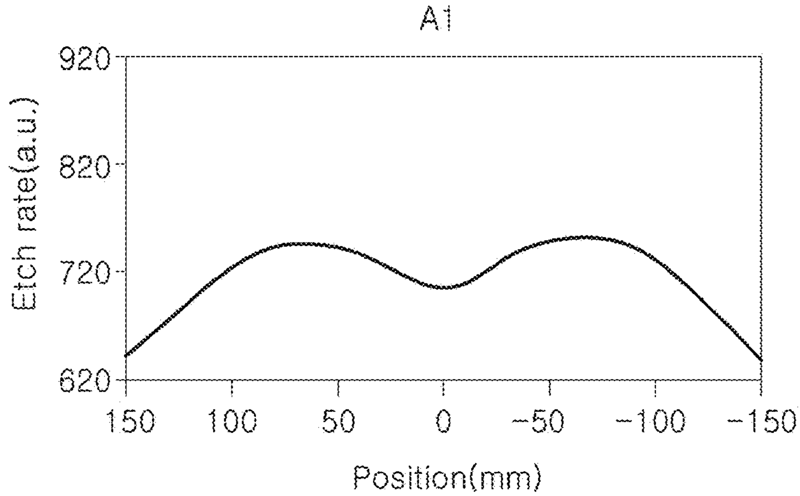


FIG. 8A

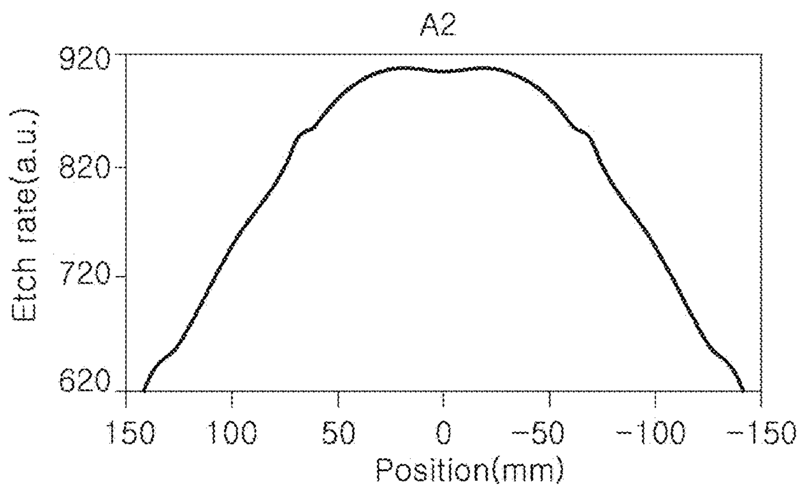


FIG. 8B

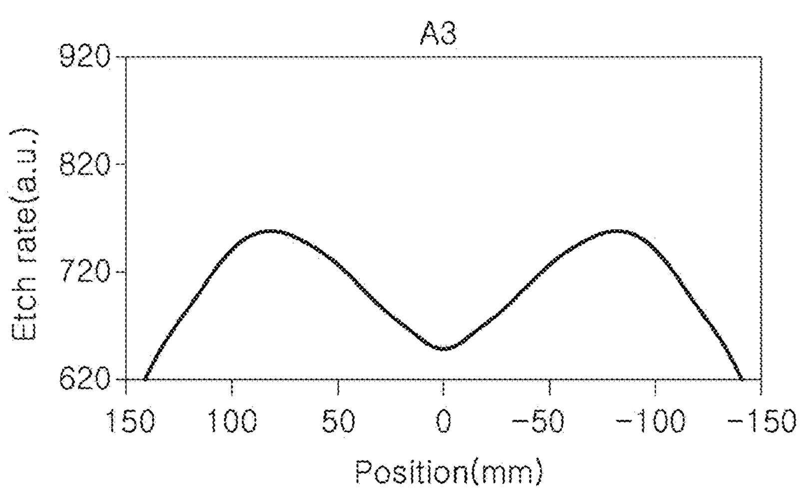
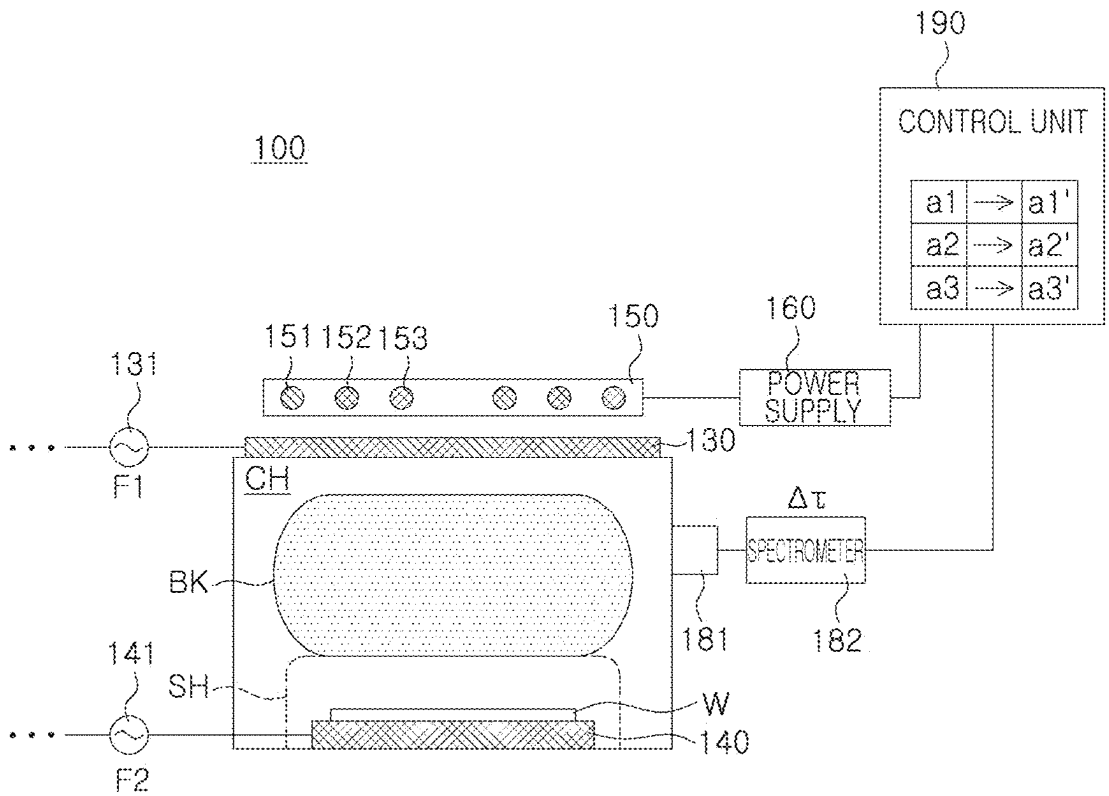


FIG. 8C



190
CONTROL UNIT
a1
a1'
a2
a2'
a3
a3'
100
150
151
152
153
160
POWER SUPPLY
131
130
F1
CH
Δτ
SPECTROMETER
BK
181
182
141
SH
W
140
F2

FIG. 9

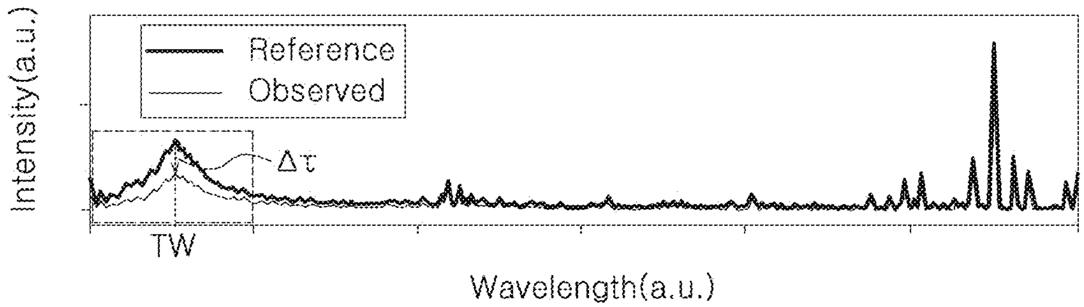


FIG. 10A

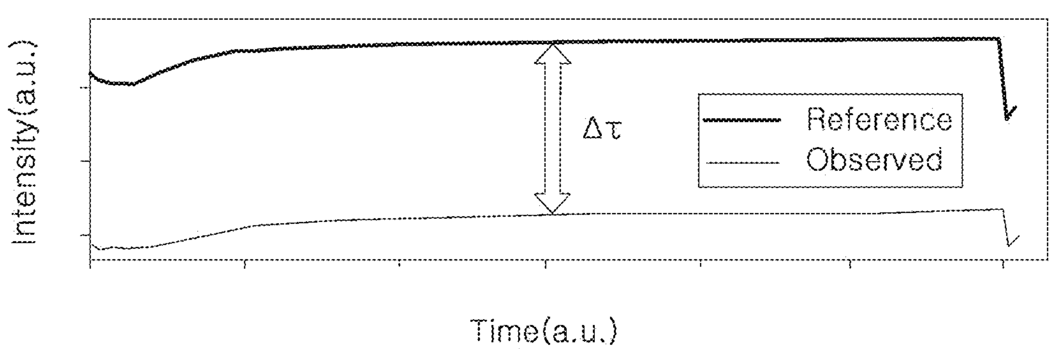


FIG. 10B

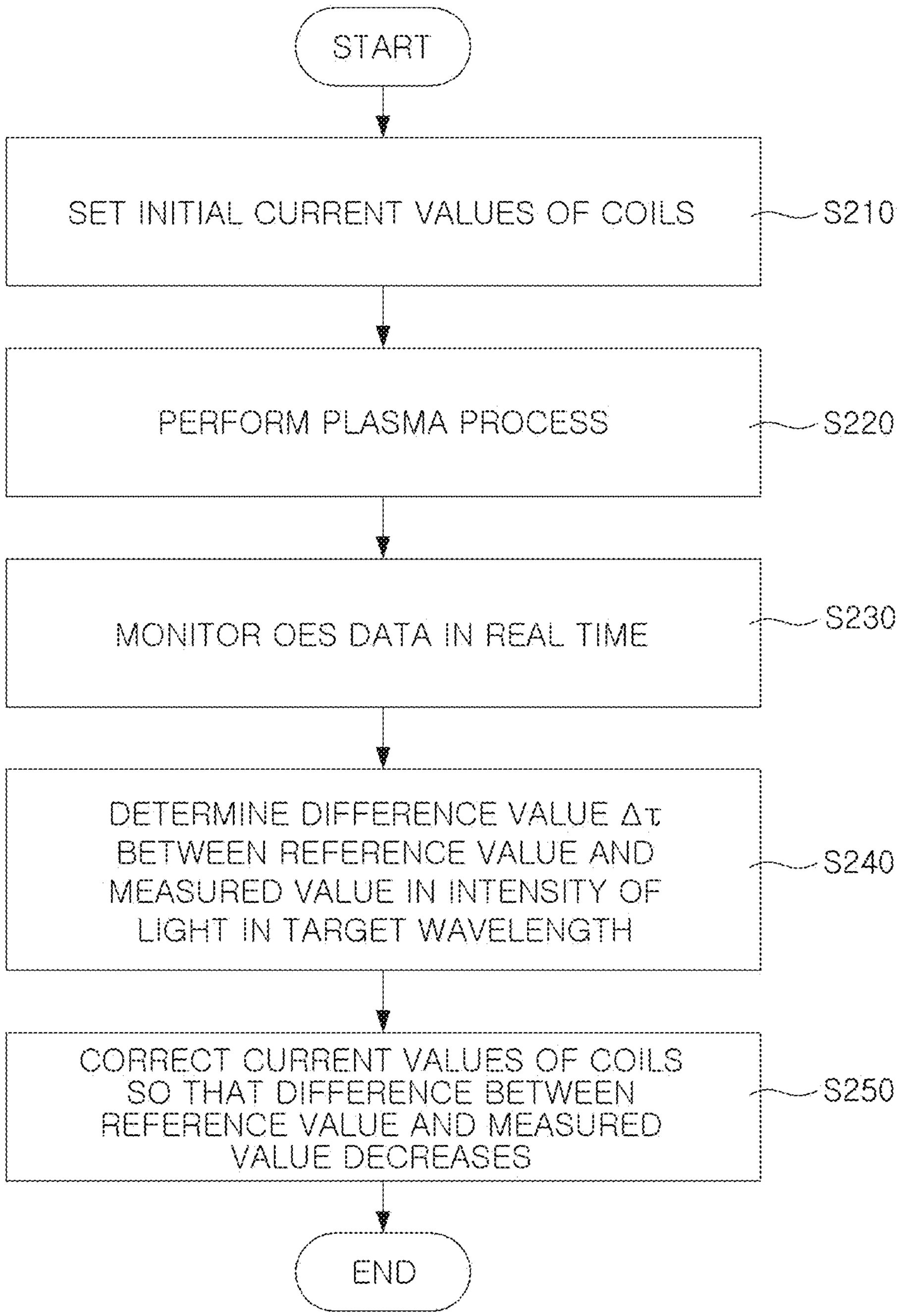

START
SET INITIAL CURRENT VALUES OF COILS
S210
PERFORM PLASMA PROCESS
S220
MONITOR OES DATA IN REAL TIME
S230
DETERMINE DIFFERENCE VALUE Δτ BETWEEN REFERENCE VALUE AND MEASURED VALUE IN INTENSITY OF LIGHT IN TARGET WAVELENGTH
S240
CORRECT CURRENT VALUES OF COILS SO THAT DIFFERENCE BETWEEN REFERENCE VALUE AND MEASURED VALUE DECREASES
S250
END

FIG. 11

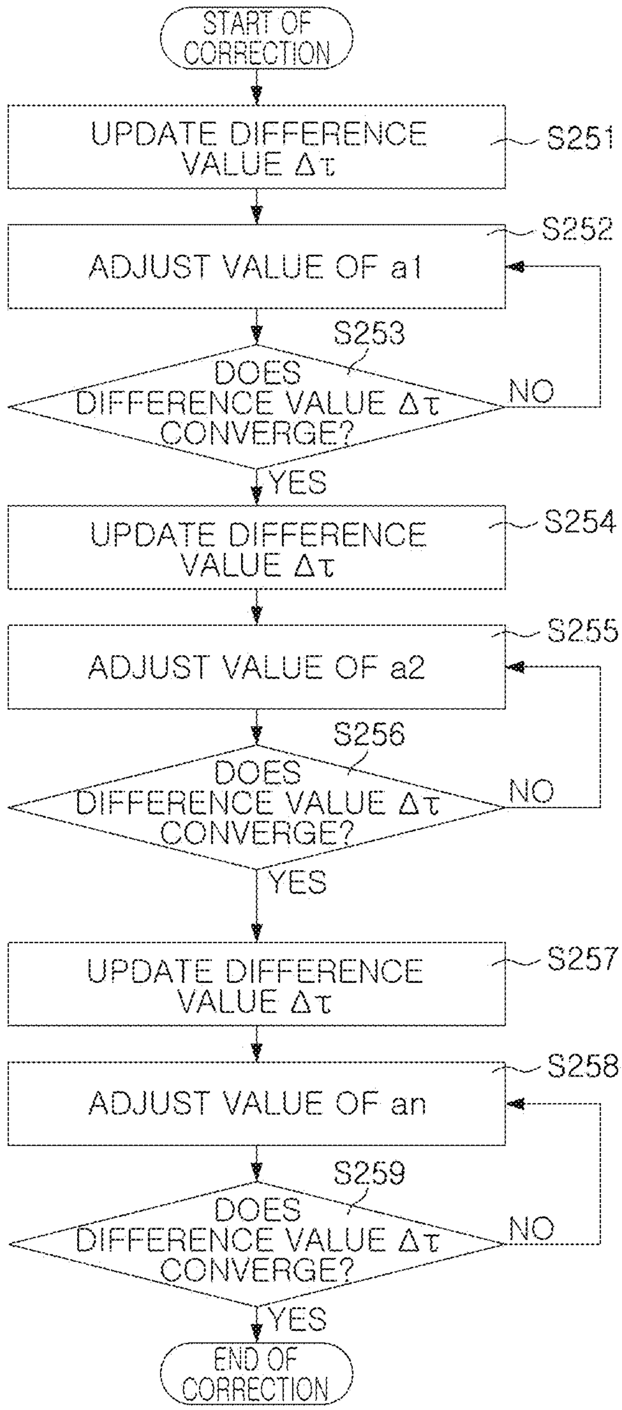

START OF CORRECTION
UPDATE DIFFERENCE VALUE Δτ
S251
ADJUST VALUE OF a1
S252
S253
DOES DIFFERENCE VALUE Δτ CONVERGE?
NO
YES
UPDATE DIFFERENCE VALUE Δτ
S254
ADJUST VALUE OF a2
S255
S256
DOES DIFFERENCE VALUE Δτ CONVERGE?
NO
YES
UPDATE DIFFERENCE VALUE Δτ
S257
ADJUST VALUE OF an
S258
S259
DOES DIFFERENCE VALUE Δτ CONVERGE?
NO
YES
END OF CORRECTION

FIG. 12

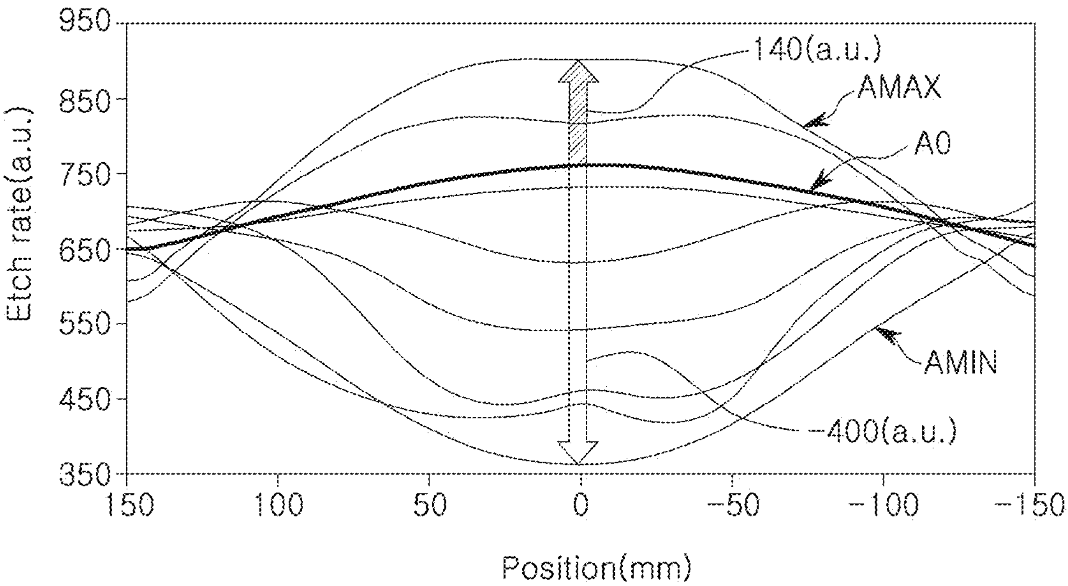

950
850
750
650
550
450
350
Etch rate(a.u.)
150
100
50
0
−50
−100
−150
Position(mm)
140(a.u.)
AMAX
A0
AMIN
−400(a.u.)

FIG. 13

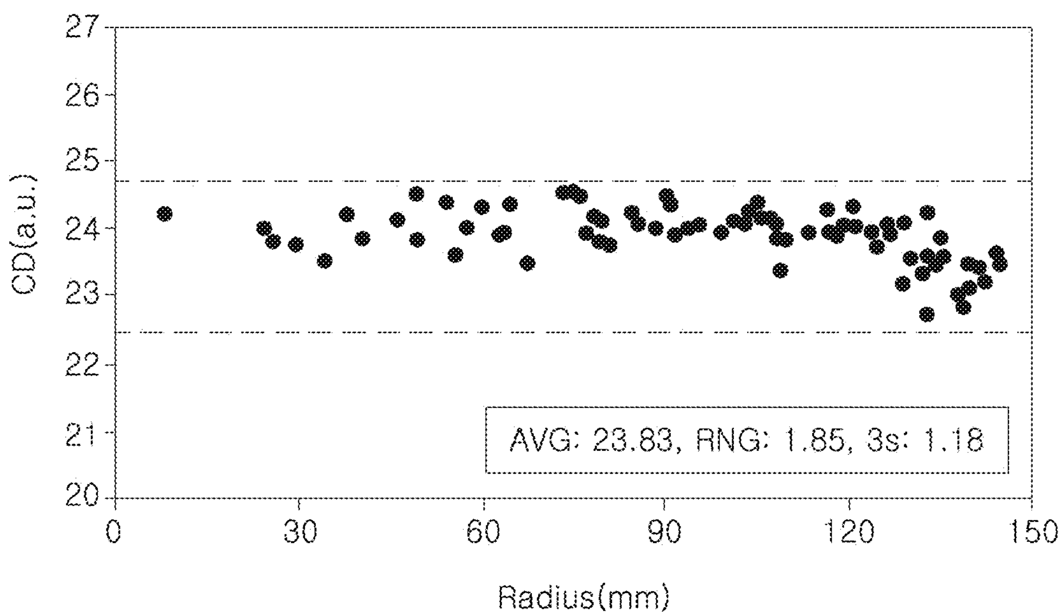

CD(a.u.)
27
26
25
24
23
22
21
20
0
30
60
90
120
150
Radius(mm)
AVG: 23.83, RNG: 1.85, 3s: 1.18

FIG. 14

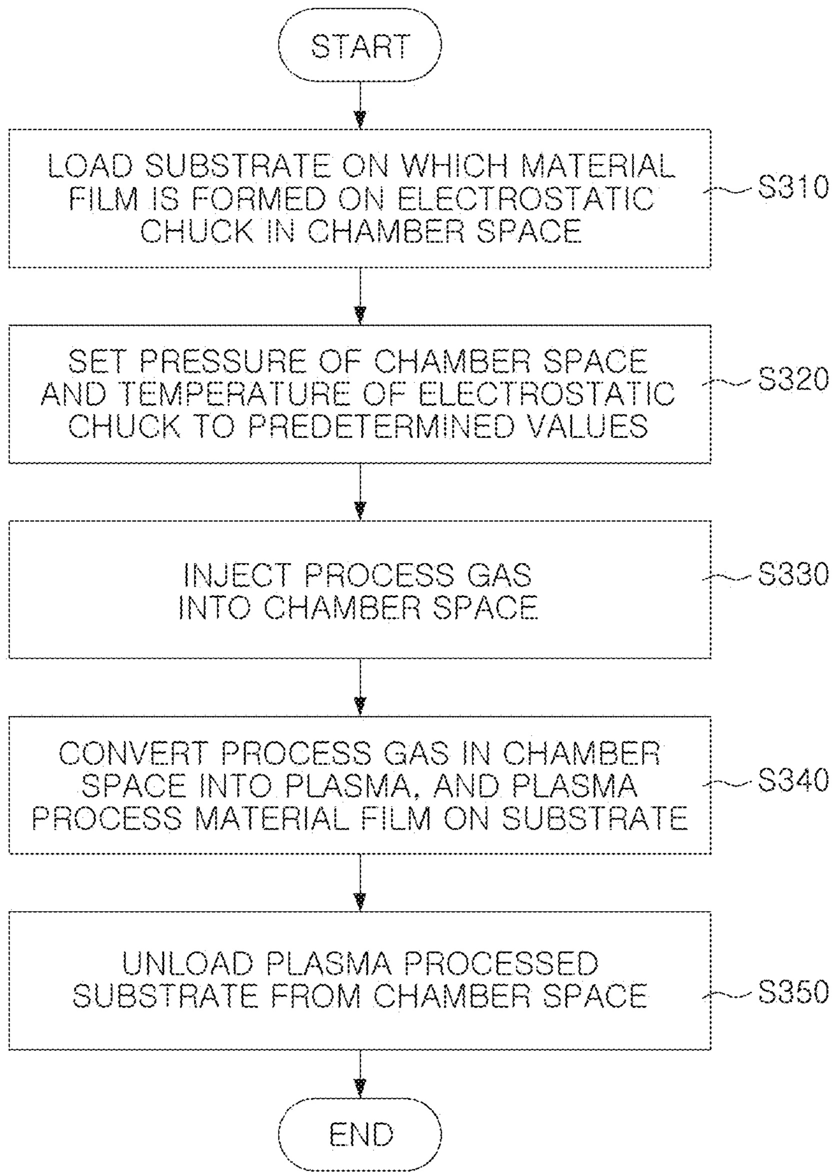

START
LOAD SUBSTRATE ON WHICH MATERIAL FILM IS FORMED ON ELECTROSTATIC CHUCK IN CHAMBER SPACE
S310
SET PRESSURE OF CHAMBER SPACE AND TEMPERATURE OF ELECTROSTATIC CHUCK TO PREDETERMINED VALUES
S320
INJECT PROCESS GAS INTO CHAMBER SPACE
S330
CONVERT PROCESS GAS IN CHAMBER SPACE INTO PLASMA, AND PLASMA PROCESS MATERIAL FILM ON SUBSTRATE
S340
UNLOAD PLASMA PROCESSED SUBSTRATE FROM CHAMBER SPACE
S350
END

FIG. 15

PLASMA PROCESSING DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of priority to Korean Patent Application No. 10-2024-0028189, filed on Feb. 27, 2024, in the Korean Intellectual Property Office, the entirety of which is incorporated herein by reference.

BACKGROUND

[0002] Semiconductor devices may be manufactured using a plasma processing device. For example, a plasma processing device may include a plasma film deposition device or a plasma etching device. In order to ensure uniform quality of manufactured semiconductor devices, it is important for the plasma processing device to uniformly control distribution of plasma in a chamber space.

SUMMARY

[0003] Some aspects of the present disclosure provide plasma processing devices capable of increasing plasma density in a region corresponding to a central portion of a substrate in a chamber space, using a magnetic field control device.

[0004] Some aspects of the present disclosure provide plasma processing devices capable of widely controlling process distribution over a wide range and uniformly controlling process distribution, using a magnetic field control device.

[0005] According to some aspects of the present disclosure, a plasma processing device includes: a chamber body defining a chamber space; an electrostatic chuck arranged to support a substrate in the chamber space; an upper electrode located at an upper portion of the chamber body; a multiple high-frequency power supply including: a first power supply configured to apply a first high-frequency power to the upper electrode, and a second power supply configured to apply a second high-frequency power to the electrostatic chuck, wherein at least one of the first high-frequency power or the second high-frequency power excites a process gas supplied to the chamber space to form a bulk plasma region and a sheath region in the chamber space, and wherein the multiple high-frequency power supply is configured to apply the at least one of the first high-frequency power or the second high-frequency power singly or simultaneously; a magnetic field control device including at least one coil located above the upper electrode, wherein the magnetic field control device is configured to form a magnetic field in the chamber space using at least one current flowing through the at least one coil; and a control unit configured to control the at least one current flowing through the at least one coil such that a magnetic flux density at a target position on a boundary of the sheath region has a value that results in (i) an electronic rotation period at the target position due to the magnetic field matching (ii) an electronic oscillation period that depends on the at least one of the first high-frequency power and the second high-frequency power.

[0006] According to some aspects of the present disclosure, a plasma processing device includes: a chamber body defining a chamber space; an electrostatic chuck arranged to support a substrate in the chamber space; an upper electrode located at an upper portion of the chamber body; a multiple high-frequency power supply including: a first power supply configured to apply a first high-frequency power to the upper electrode, and a second power supply configured to apply a second high-frequency power to the electrostatic chuck, wherein at least one of the first high-frequency power or the second high-frequency power excites a process gas supplied to the chamber space to form a bulk plasma region and a sheath region in the chamber space, and wherein the multiple high-frequency power supply is configured to apply the at least one of the first high-frequency power or the second high-frequency power singly or simultaneously; an optical interface configured to receive light from the chamber space; a spectrometer configured to monitor a light intensity at a target wavelength of the light; a magnetic field control device including at least one coil located above the upper electrode, wherein the magnetic field control device is configured to form a magnetic field in the chamber space using at least one current flowing through the at least one coil; and a control unit configured to: set at least one initial value of the at least one current, cause a plasma process to be performed on the substrate, and based on a measured value of the light intensity at the target wavelength being different from a reference value, adjust the at least one current such that a difference between the measured value and the reference value decreases.

[0007] According to some aspects of the present disclosure, a plasma processing device includes: a chamber body defining a chamber space; an electrostatic chuck arranged to support a substrate in the chamber space; an upper electrode located at an upper portion of the chamber body; a multiple high-frequency power supply including: a first power supply configured to apply a first high-frequency power to the upper electrode, and a second power supply configured to apply a second high-frequency power to the electrostatic chuck, wherein at least one of the first high-frequency power or the second high-frequency power excites a process gas supplied to the chamber space to form a bulk plasma region and a sheath region in the chamber space, and wherein the multiple high-frequency power supply is configured to apply the at least one of the first high-frequency power or the second high-frequency power singly or simultaneously; a magnetic field control device including at least one coil located above the upper electrode, wherein the magnetic field control device is configured to form a magnetic field in the chamber space using at least one current flowing through the at least one coil; and a control unit configured to: acquire a distribution map indicating a process value as a function of a distance from a center of the substrate, and based on when the process value decreasing toward the center of the substrate in the distribution map, determining at least one value of the at least one current flowing through the at least one coil such that magnetic resonance occurs at a target position at which a boundary of the sheath region and a central axis of the chamber space meet.

BRIEF DESCRIPTION OF DRAWINGS

[0008] The above and other aspects, features, and advantages will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:

[0009] FIG. **1** is a diagram illustrating an example of a plasma processing device.

[0010] FIG. **2** is a top view illustrating an example of a magnetic field control device.

[0011] FIG. **3** is a side view illustrating movement of electrons due to a magnetic field in an example of a plasma processing device.

[0012] FIG. **4** is a chart illustrating an example of process distribution of a plasma processing device.

[0013] FIG. **5** is a diagram illustrating an example of a method of controlling a magnetic field of a plasma processing device.

[0014] FIG. **6** is a flowchart illustrating an example of a method of controlling a magnetic field.

[0015] FIGS. **7**A to **7**C are charts illustrating simulation results of magnetic field distribution.

[0016] FIGS. **8**A to **8**C are charts illustrating etching rates as functions of position.

[0017] FIG. **9** is a diagram illustrating an example of a method of controlling a magnetic field of a plasma processing device.

[0018] FIGS. **10**A and **10**B are charts illustrating optical emission spectroscopy (OES) monitoring.

[0019] FIG. **11** is a flowchart illustrating an example of a method of controlling a magnetic field.

[0020] FIG. **12** is a flowchart illustrating an example of a method of correcting current values.

[0021] FIG. **13** is a chart illustrating various distribution maps associated with a method of controlling a magnetic field.

[0022] FIG. **14** is a chart illustrating an example of process distribution.

[0023] FIG. **15** is a flowchart illustrating an example of a plasma processing method.

DETAILED DESCRIPTION

[0024] In the following description, like reference numbers refer to like elements having the same characteristics, except where noted otherwise or suggested otherwise by context.

[0025] FIG. **1** is a diagram illustrating an example of a plasma processing device. Referring to FIG. **1**, a plasma processing device **100** includes a chamber body **110**, a gas supply device **120**, an upper electrode **130**, a first power supply **131**, a first impedance adapter **132**, an electrostatic chuck **140**, a second power supply **141**, a second impedance adapter **142**, an electromagnet **150**, a third power supply **160**, an exhaust device **170**, an optical interface **181**, a spectrometer **182**, and a control unit **190**.

[0026] The chamber body **110** may serve as a housing forming a chamber space CH (labeled, for example, in FIG. **3**) defined by an external wall. The chamber space CH may be used to perform a plasma process of processing a substrate W to be processed, using plasma PLA generated by exciting a process gas supplied by the gas supply device **120**. The external wall may be formed of a material having excellent wear resistance and excellent corrosion resistance. The chamber body **110** may maintain the chamber space CH in a sealed state having a given pressure and a given temperature during the plasma process, for example, an etching process. The exhaust device **170** may be disposed on the external wall of the chamber body **110** to exhaust gas from an internal space such as the chamber space CH.

[0027] The gas supply device **120** may supply the process gas for performing the plasma process to the chamber space CH. For example, the process gas may include a gas for physical etching, such as argon (Ar) or the like, a gas for chemical etching, such as C_xF_y or the like, or the like.

[0028] The upper electrode **130** may be disposed in an upper portion of the chamber body **110**. A first high-frequency power source, for example, an RF power source, may be applied to the upper electrode **130** by the first power supply **131**.

[0029] The electrostatic chuck **140** may be disposed in the chamber space CH, and may fix the substrate W on an upper surface thereof using static electricity. A second high-frequency power source, for example, an RF power source, may be applied to the electrostatic chuck **140** by the second power supply **141**. For example, the electrostatic chuck **140** may function as a lower electrode.

[0030] The process gas supplied from the gas supply device **120** may be converted into a plasma PLA state by the first high-frequency power source applied exclusively to the upper electrode **130**, by the second high-frequency power source applied exclusively to the lower electrode, or by the first high-frequency power source and the second high-frequency power source that may be applied to the upper electrode **130** and the lower electrode simultaneously.

[0031] Positive ions included in the plasma PLA may be accelerated by the second high-frequency power source, and the plasma process may be performed by irradiating the accelerated ions to the substrate W. The first power supply **131** and the second power supply **141**, capable of applying a power source to any one of the upper electrode **130** and the lower electrode alone or simultaneously to the upper electrode **130** and the lower electrode, may also be referred to as a multiple high-frequency power supply.

[0032] Plasma may include electrons, positive ions, and neutrons. The chamber space CH containing the plasma PLA may include a bulk plasma region BK having high-density plasma and a sheath region SH mainly containing positive ions and neutrons.

[0033] As illustrated in FIG. **1**, in the plasma processing device **100**, plate electrodes as the upper electrode **130** and the lower electrode may be arranged to face each other, and the plasma processing device **100** may perform the plasma process using plasma (charge coupled plasma (CCP)) generated by capacitance coupling.

[0034] In some implementations, the first impedance adapter **132** for minimizing reflected power by matching impedance of a circuit and a power source is connected between the first power supply **131** and the upper electrode **130**. Likewise, the second impedance adapter **142** may be connected between the second power supply **141** and the electrostatic chuck **140**.

[0035] The electromagnet **150** may be disposed on the upper electrode **130**, and may form a magnetic field in the chamber space CH. The magnetic field formed in the chamber space CH may control a density of the plasma by controlling movement of the electrons included in the plasma. The electromagnet **150** may include a plurality of coils **151**, **152**, and **153** forming the magnetic field.

[0036] The third power supply **160** may supply a power source to the plurality of coils **151**, **152**, and **153** such that a current flows through each of the plurality of coils **151**, **152**, and **153**. The third power supply **160** may form a static magnetic field in the chamber space CH by supplying a static power source to the plurality of coils **151**, **152**, and **153**. The electromagnet **150** and the third power supply **160** may be collectively referred to as a magnetic field control device.

[0037] The exhaust device **170** may depressurize the chamber space by discharging the process gas in the chamber space CH externally. For example, the exhaust device **170** may include a pump device.

[0038] The optical interface **181** may acquire or receive light output from the chamber space CH for process monitoring. For example, the optical interface **181** may be referred to as a view port. In addition, the light may include a light component of a specific wavelength emitted when atoms in an excited state in the plasma PLA is relocated to a lower energy level.

[0039] The spectrometer **182** may analyze light acquired from the optical interface **181**. For example, the spectrometer **182** may perform optical emission spectroscopy (OES) analysis to separate the light acquired from the optical interface **181** by wavelength, and may perform process monitoring, based on the light component of specific wavelength(s).

[0040] The control unit **190** may control various operations such as the gas supply device **120**, the first power supply **131**, the second power supply **141**, the third power supply **160**, the exhaust device **170**, the spectrometer **182**, and the like. The control unit **190** can include digital and/or analog circuitry, and can include a computing device configured to perform the control operations described herein.

[0041] To have uniform quality of a semiconductor device manufactured on the substrate W by the plasma process, uniform process distribution is required. Process distribution may refer to distribution of semiconductor process result values. For example, the process distribution may include distribution of an etching rate for each region of the substrate W, distribution of a critical dimension (CD) for each region of the substrate W, or the like.

[0042] When the plasma distribution for each region of the chamber space CH is not uniform, the process distribution for each region of the substrate W on which the plasma process was performed may also become non-uniform. Therefore, in order for the process distribution in each region of the substrate W to be uniform, the plasma distribution in each region of the chamber space CH may be precisely controlled.

[0043] The plasma processing device **100** may control plasma distribution for each region of the chamber space CH by forming a magnetic field in the chamber space CH using the electromagnet **150**. A Lorentz force caused by the magnetic field may act from a central portion of the chamber space CH toward a peripheral portion of the chamber space CH, and electrons may move to the peripheral portion of the chamber space CH. Therefore, when controlling the plasma distribution based only on the Lorentz force, the plasma distribution may be controlled only in a manner in which plasma density in the central portion of the chamber space CH decreases, and it may be difficult to uniformly control the plasma density.

[0044] According to some implementations, the plasma processing device **100** increases the efficiency of generating plasma in the central portion of the chamber space CH by generating magnetic resonance in the central portion. Accordingly, the plasma processing device **100** may control plasma distribution in a manner in which plasma density in the central portion of the chamber space CH increases, and as a result, a control range of plasma distribution may be improved. In addition, the plasma processing device **100** may uniformly control the process distribution even when the process result value in the central portion of the chamber space CH has low process distribution.

[0045] Hereinafter, a structure of the magnetic field control device will be described in detail with reference to FIGS. **2** to **4**. Plasma density control by the Lorentz force will be first described.

[0046] As shown in FIG. **2**, a magnetic field control device **200** may include an electromagnet **150** and a third power supply **160**, as described with reference to FIG. **1**.

[0047] The electromagnet **150** may include a plurality of coils **151**, **152**, and **153**. The plurality of coils **151**, **152**, and **153** may be arranged along concentric circles on an X-Y plane centered on a central axis passing through a center of the substrate W in a third direction Z. The central axis may, though need not, coincide with a central axis of a chamber space CH described with reference to FIG. **1**, and may further, though need not, pass through a center of the upper electrode **130** and a center of the electrostatic chuck **140**.

[0048] In some implementations, the electromagnet **150** includes a cylindrical yoke in the plurality of coils **151**, **152**, and **153**. The electromagnet **150** may further include a mold portion supporting the plurality of coils **151**, **152**, and **153**.

[0049] In the example of FIG. **2**, the plurality of coils **151**, **152**, and **153** are illustrated as partially open annular shapes, but, in some implementations, the plurality of coils **151**, **152**, and **153** may be implemented as spring type coils wound around the central axis.

[0050] The third power supply **160** may independently control currents a**1**, a**2**, and a**3** flowing through the plurality of coils **151**, **152**, and **153**. For example, the third power supply **160** may include a first power source **161** applying a first current a**1** to a first coil **151**, a second power source **162** applying a second current a**2** to a second coil **152**, and a third power source **163** applying a third current a**3** to a third coil **153**.

[0051] When the currents a**1**, a**2**, and a**3** are applied to the plurality of coils **151**, **152**, and **153**, a magnetic field may be formed around the plurality of coils **151**, **152**, and **153**. A Lorentz force may act on electrons included in plasma PLA by the currents a**1**, a**2**, and a**3** flowing in the plurality of coils **151**, **152**, and **153** and the magnetic field formed in the chamber space CH disposed below the plurality of coils **151**, **152**, and **153**, and a distribution of the plasma PLA may be controlled as the electrons move by the Lorentz force.

[0052] FIG. **3** is a view illustrating movement of electrons due to a magnetic field in a plasma processing device. As shown in FIG. **3**, the plasma processing device **100** includes a chamber space CH, an upper electrode **130**, an electrostatic chuck **140**, a substrate W, and an electromagnet **150** in a plasma processing device **100**, having characteristics as described with reference to FIG. **1**. To control electrons e^- in the chamber space CH, the plasma processing device **100** may apply a magnetic field toward the chamber space CH by applying a current to a plurality of coils **151**, **152**, and **153**. FIG. **3** illustrates a direction of an electric field E in a cross-section of a third coil **153**.

[0053] Due to the current applied to the plurality of coils **151**, **152**, and **153**, the magnetic field may be formed around the plurality of coils **151**, **152**, and **153**. The magnetic field may be formed in a direction perpendicular to the electric field E, and may be formed to surround the plurality of coils **151**, **152**, and **153**.

[0054] FIG. **3** illustrates directions of some components of a magnetic field B. In the magnetic field B, directional components spreading from a central axis of a chamber to a peripheral portion may be dominant, as illustrated in FIG. **3**.

When the electric field E and the magnetic field B in directions perpendicular to each other are applied to the electrons e^-, the Lorentz force may act on the electrons e^- in a direction corresponding to the cross product of the electric field E and magnetic field B. Electron drift may occur in which electrons e^-, driven by the Lorentz force, move while rotating in a spiral shape, based on a direction of the magnetic field B. The electron drift may be referred to as E×B drift. Due to E×B drift, the electrons e^- may be concentrated in a peripheral portion of the chamber space CH, and positive ions may also move along with the electrons e.

[0055] As a result, when distribution of plasma PLA is controlled based only on the Lorentz force, the distribution of plasma PLA may be controlled in a manner such that plasma density in the peripheral portion of the chamber space CH is strengthened and plasma density in a central portion of the chamber space CH is weakened.

[0056] When the distribution of plasma PLA is controlled in only one manner, as described with reference to FIG. **3**, it may be difficult to uniformly control process distribution in an environment having various process distributions, depending on the process environment.

[0057] For example, FIG. **4** illustrates an example of process distribution of a plasma processing device. The graph in FIG. **4** illustrates distribution of critical dimension (CD) according to a radius from a central axis of a substrate W. The CD in FIG. **4** represents CD when a magnetic field is not formed by a magnetic field control device.

[0058] For example, a position having a radius of '0' represents a center of the substrate W, and a position having a radius of '150' represents an edge of the substrate W. Referring to FIG. **4**, in process distribution, the CD may decrease toward the center of the substrate W, and the CD may increase toward the edge of the substrate W. The CD may have a positive correlation with an etching rate. For example, in a region not sufficiently etched, a width between adjacent patterns may not be wide enough. In order for the process distribution to be even, the CD may need to be adjusted upward by adjusting the etching rate in a central portion of the substrate W upward.

[0059] When a density of the plasma PLA is controlled only depending on Lorentz force, as described with reference to FIG. **3**, control may be limited to decreasing the density of the plasma PLA in a central portion of the chamber space CH and to increasing the density of the plasma PLA in a peripheral portion of the chamber space CH. Therefore, it may be difficult to evenly correct the process distribution illustrated in FIG. **4**.

[0060] According to some implementations of the present disclosure, a magnetic field control device is not limited to controlling the density of plasma PLA by controlling movement of electrons by applying Lorentz force to electrons in the chamber space CH. The density of plasma PLA may be controlled by increasing efficiency of plasma generation by generating magnetic resonance in the central portion of the chamber space CH, to generate plasma PLA in the central portion.

[0061] Hereinafter, methods of controlling a magnetic field of a plasma processing device according to some implementations will be described in detail with reference to FIGS. **5** to **8**.

[0062] As shown in FIG. **5**, a plasma processing device **100** includes a chamber space CH, an upper electrode **130**, a first power supply **131**, an electrostatic chuck **140**, a second power supply **141**, a substrate W, an electromagnet **150**, and a third power supply **160**, as described with reference to FIG. **1**.

[0063] The first power supply **131** may provide a first high-frequency power source having a first frequency F**1** to the upper electrode **130**, and the second power supply **141** may provide a second high-frequency power source having a second frequency F**2** to the electrostatic chuck **140**.

[0064] As explained with reference to FIG. **1**, plasma PLA may be formed by exciting a process gas by the first high-frequency power source applied to the upper electrode **130** or the second high-frequency power source applied to a lower electrode, either alone or simultaneously. A plasma process may be performed on the substrate W by accelerating positive ions included in the plasma PLA by the second high-frequency power source.

[0065] Plasma may include electrons, positive ions, and neutrons. The electrons may be much lighter than the positive ions, and may have higher mobility than the positive ions. Therefore, the electrons included in the plasma may move to the electrostatic chuck **140** faster than the positive ions, thereby charging the electrostatic chuck **140** as a negative electrode and pushing out the electrons. As a result, the positive ions and the neutrons mainly remain around the electrostatic chuck **140**, and a region mainly containing the positive ions and the neutrons may be referred to as a sheath region SH. Due to the positive ions in the sheath region SH and the electrons in the electrostatic chuck **140**, a voltage may rapidly drop in the sheath region SH. An electric field formed in the sheath region may accelerate the positive ions in the sheath region, and may generate high kinetic energy. In some implementations, an etching process is performed on the substrate W when the positive ions having high energy collide with the substrate.

[0066] A bulk plasma region BK may include electrons, positive ions, and neutrons. The electrons in the bulk plasma region BK may oscillate at a frequency determined by the first high-frequency power source and the second high-frequency power source. For example, the electrons may oscillate at the first frequency F**1**, which may be a frequency of the first high-frequency power source for exciting the process gas, or at the second frequency F**2**, which may be a frequency of the second high-frequency power source. Additionally, a frequency at which electrons oscillate may further include a common multiple of the first frequency F**1** and the second frequency F**2**.

[0067] Coils **151**, **152**, and **153** included in the electromagnet **150** may form a magnetic field in a chamber space CH by currents a**1**, a**2**, and a**3** applied from the third power supply **160**. FIG. **5** illustrates a distribution of the magnetic field formed in the chamber space CH according to a magnetic flux density. In the chamber space CH, a region illustrated in relatively dark shade has a relatively high magnetic flux density, and a region illustrated in relatively light shade has a relatively low magnetic flux density. Magnetic flux lines representing positions with the same magnetic flux density in the chamber space CH are illustrated in the form of contour lines. In FIG. **5**, an indication of the magnetic field that may be formed in the sheath region SH is omitted.

[0068] Referring to FIG. **5**, the magnetic flux density of the magnetic field formed in the chamber space CH varies based on position. The magnetic flux density of the magnetic field may be changed depending on a magnitude and a direction of the currents a**1**, a**2**, and a**3**, but may be generally higher toward an upper portion of the chamber space CH and toward a central axis of the chamber space CH.

[0069] As explained with reference to FIG. **3**, the electrons included in the plasma PLA may rotate by the Lorentz force due to currents applied to the coils **151**, **152**, and **153** and magnetic fields formed by the coils **151**, **152**, and **153**. A rotation period of the electrons may be determined depending on the magnetic flux density of the magnetic field. For example, the rotation period of the electrons may be changed depending on a position of the chamber space CH.

[0070] When an oscillation period of the electrons based on the first high-frequency power source matches a rotation period of the electrons based on the magnetic flux density of the magnetic field, magnetic resonance may occur. For example, an amplitude of electrons oscillating at a certain period may increase significantly, as the electrons continuously receive an external force having the same frequency as the oscillation period. Therefore, the average kinetic energy of the electrons may increase due to the magnetic resonance. When the kinetic energy of electrons increases, temperatures of the electrons may increase, increasing the efficiency of generating plasma (efficiency of plasma generation).

[0071] Since the rotation period of the electrons varies depending on a position of the chamber space CH, magnetic resonance may occur in a local region in the chamber space CH in which the rotation period and the oscillation period of the electrons match, and the efficiency of generating plasma in this region may increase. In FIG. **5**, example magnetic flux lines are illustrated as bold lines representing positions with magnetic flux densities M in which magnetic resonance may occur.

[0072] As illustrated in FIG. **5**, when magnetic resonance occurs in a position in which a central axis of the chamber space CH passing through a center of the substrate W in a vertical direction meets a boundary of the sheath region SH, a density of plasma in this position may increase, and when performing a plasma process, a large amount of positive ions may be irradiated to/directed to the center of the substrate W. As a result, process result values at the center of the substrate W may be controlled upward.

[0073] According to some implementations, the plasma processing device **100** controls the currents a**1**, a**2**, and a**3** to have a magnetic flux density in which the rotation period of the electrons matches the oscillation period of the electrons, at a distance R ranging from centers of the coils **151**, **152**, and **153** to a position in which the boundary of the sheath region SH meets, based on a thickness S of the sheath region.

[0074] According to some implementations, the plasma processing device **100** controls upward process result values in a central portion of the substrate W by increasing the density of plasma at a target position at which the central axis of the chamber space CH and the boundary of the sheath region SH meet. Therefore, a control range of process distribution may be expanded, and it may be more easy to adjust the process distribution to be uniform for each region of the substrate W.

[0075] FIG. **6** is a flowchart illustrating a method of controlling a magnetic field. The method of controlling the magnetic field may include S**110** to S**140**, and S**110** to S**140** may be performed by a control unit **190**, as described with reference to FIG. **1**.

[0076] In S**110**, the control unit **190** may determine a magnetic flux density M at which electrons may cause or experience magnetic resonance, based on an oscillation period of the electrons. As explained with reference to FIG. **5**, the oscillation period of the electrons may be influenced by a first high-frequency power source and/or a second high-frequency power source. For example, the oscillation period of the electrons may have at least one of a first frequency F**1** component, a second frequency F**2** component, or a common multiple of a first frequency F**1** and a second frequency F**2**. In addition, a rotation period of the electrons may be determined based on a magnetic flux density of a magnetic field, and the rotation period of the electrons may be derived from the well-known Lorentz force formula. The control unit **190** may determine a magnetic flux density M at which the rotation period of the electrons matches the oscillation period of the electrons.

[0077] In S**120**, the control unit **190** may determine a thickness S of a sheath region. The thickness S of the sheath region may mean a distance ranging from an upper surface of an electrostatic chuck **140** to a boundary of the sheath region along a central axis of the chamber space CH or a central axis of the substrate W. The thickness S of the sheath region may be calculated based on process conditions, such as at least one of a temperature of the chamber space CH, a potential difference between the boundary of the sheath region SH and the electrostatic chuck **140**, a pressure of the chamber space CH, or a flow rate of a process gas.

[0078] In some implementations, the control unit **190** adjusts calculated values of the thickness S of the sheath region, based on process results. For example, the control unit **190** may control currents a**1**, a**2**, and a**3**, based on the calculated values, and may adjust the calculated values of the thickness S of the sheath region, based on results of performing a plasma process.

[0079] In S**130**, the control unit **190** may calculate a current value set A capable of forming a magnetic flux density M capable of causing magnetic resonance at a distance R that is the distance from a central portion of the coils to the boundary of the sheath region SH. For example, using the Biot-Savart law, a magnetic flux density for a position may be calculated based on (i) a distance of the position from the coils **151**, **152**, and **153**, and (ii) the currents a**1**, a**2**, and a**3** applied to coils **151**, **152**, and **153**. The control unit **190** may calculate a current value set A, which may be a combination of values of currents a**1**, a**2**, and a**3** capable of forming a magnetic flux density M at the distance R ranging from central portions of the coils to the boundary of the sheath region SH (for example, at the boundary of the sheath region SH and along the central axis), using the Biot-Savart law

[0080] In S**140**, the control unit **190** may control a third power supply **160** such that the currents a**1**, a**2**, and a**3** flowing through the coils **151**, **152**, and **153** have current values included in the current value set A, to generate a magnetic field that causes magnetic resonance at the boundary of the sheath region SH (e.g., at the boundary of the sheath region SH along the central axis).

[0081] FIGS. **7**A to **7**C illustrate magnetic field distribution according to a position in a chamber space when currents are applied to coils using different sets of current values. In the graphs of FIGS. **7**A to **7**C, the horizontal axis represents a distance (radius) from a central axis of a chamber space CH. The vertical axis represents a height in the chamber space CH, in reference to an upper surface of an electrostatic chuck. On the vertical axis, a thickness S of a sheath region is displayed, based on the upper surface of the electrostatic chuck.

[0082] FIG. **7**A illustrates magnetic field distribution when a first current value set A**1** is applied to the coils. Regions illustrated in a relatively dark shade in the graph have a relatively high magnetic flux density, and regions illustrated in a relatively light shade have a relatively low magnetic flux density. Magnetic flux lines representing positions having the same magnetic flux density are illustrated. In particular, a magnetic flux line illustrated as a thick line illustrates a position of the chamber space CH having a magnetic flux density in which a oscillation period of electrons and a rotation period of the electrons match for oscillation periods of 13 MHz, 40 MHZ, 60 MHz, and 120 MHz.

[0083] FIG. **7**B illustrates magnetic field distribution when a second current value set A**2** is applied to the coils. In FIG. **7**B, shades and magnetic flux lines are illustrated according to a magnetic flux density, like FIG. **7**A. In particular, a magnetic flux line illustrated as a thick line illustrates a position of the chamber space CH having a magnetic flux density in which a oscillation period of electrons and a rotation period of the electrons match for oscillation periods of 13 MHz, 40 MHz, 60 MHz, and 120 MHz.

[0084] FIG. **7**C illustrates magnetic field distribution when a third current value set A**3** is applied to the coils. In FIG. **7**C, shades and magnetic flux lines according to a magnetic flux density are illustrated, similar to FIGS. **7**A and **7**B. In particular, a magnetic flux line illustrated as a thick line illustrates a position of the chamber space CH having a magnetic flux density in which a oscillation period of electrons and a rotation period of the electrons match for oscillation periods of 40 MHz, 60 MHz, and 120 MHz.

[0085] The oscillation period of the electrons may be determined according to a frequency of a high-frequency power source applied to at least one of an upper electrode **130** or a lower electrode. For example, when a first high-frequency power source having a first frequency F**1** is applied alone to the upper electrode **130**, the oscillation period of the electrons may include a first frequency F**1** component. Also, when a second high-frequency power source having a second frequency F**2** is applied alone to the lower electrode including the electrostatic chuck **140**, the oscillation period of the electrons may include a second frequency F**2** component. When the first high-frequency power source and the second high-frequency power source are applied simultaneously, the oscillation period of the electrons may include the first frequency F**1** component, the second frequency F**2** component, and a common multiple of the first frequency F**1** and the second frequency F**2**.

[0086] For example, when the first frequency F**1** is 60 MHz and the second frequency F**2** is 40 MHz, and the first high-frequency power source and the second high-frequency power source are applied at the same time, the oscillation period of the electrons may include 60 MHz, 40 MHZ, and a common multiple of 120 MHz. When a magnetic flux line corresponding to 60 MHz, a magnetic flux line corresponding to 40 MHz, or a magnetic flux line corresponding to 120 MHz passes through a target position in which a central portion of the chamber space CH and a boundary S of the sheath region meet, magnetic resonance may occur at the target position.

[0087] In examples of FIGS. **7**A and **7**C, since none of a magnetic flux line corresponding to 60 MHz, a magnetic flux line corresponding to 40 MHz, and a magnetic flux line corresponding to 120 MHz pass through the target position at the boundary of the sheath region, magnetic resonance does not occur at the target position. In an example of FIG. **7**B, a magnetic flux line corresponding to 60 MHz passes through the target position, and magnetic resonance occurs at the target position.

[0088] Therefore, based on the oscillation period of the electrons determined by at least one of the first high-frequency power source or the second high-frequency power source, the second current value set A**2** may be selected such that magnetic resonance occurs at a position at which the central portion of the chamber space CH and the boundary S of the sheath region meet.

[0089] FIGS. **8**A to **8**C illustrate an etching rate according to a position of a substrate W when a first high-frequency power source having a frequency of 60 MHz is applied alone to an upper electrode **130** and currents are applied to coils using different current value sets A**1**, A**2**, and A**3**, as illustrated in FIGS. **7**A to **7**C. In graphs of FIGS. **8**A to **8**C, a horizontal axis represents a relative position from a central axis of the substrate W, and a vertical axis represents the etching rate. On the horizontal axis, '0' may represent a central portion of the substrate W, and '−150' and '150' may represent both end portions of the substrate in any direction, parallel to an upper surface of the substrate.

[0090] FIG. **8**A illustrates etching rate distribution when a first current value set A**1**, as described with reference to FIG. **7**A, is applied to the coils. Referring to FIG. **7**A, a magnitude of a magnetic field formed by the coils may not be sufficient to cause magnetic resonance on a boundary of a sheath region, and a magnetic flux line corresponding to 60 MHz passes through a position spaced approximately 40 mm away from an electrostatic chuck **140** in a central portion of a chamber space CH. Plasmas generated due to magnetic resonance in a position spaced apart from the boundary S of the sheath region may move to a peripheral portion of the substrate W due to E×B drift, and may participate less in plasma processing of the central portion of the substrate W. Therefore, in the substrate W, an etching rate of the central portion may be lower than an etching rate of the peripheral portion.

[0091] FIG. **8**B illustrates etching rate distribution when a second current value set A**2**, as described with reference to FIG. **7**B, is applied to the coils. Referring to FIG. **7**B, a magnitude of a magnetic field formed by the coils is appropriate to generate magnetic resonance at a target position at which a central portion of a chamber space CH and a boundary S of a sheath region meet. For example, a magnetic flux line corresponding to 60 MHz passes through the target position. A large amount of plasma generated due to magnetic resonance at the target position may actively participate in plasma processing of the central portion of the substrate W, and, in the substrate W, an etching rate of the central portion may be higher than an etching rate of the peripheral portion.

[0092] FIG. **8**C illustrates etching rate distribution when a third current value set A**3**, as described with reference to FIG. **7**C, is applied to the coils. Referring to FIG. **7**C, a magnitude of a magnetic field formed by the coils is excessive to cause magnetic resonance in a position at which a central portion of a chamber space CH and a boundary S of a sheath region meet, and a magnetic flux line corresponding to 60 MHz passes through a position in which the boundary of the sheath region S and a peripheral portion of the chamber space CH meet. Therefore, magnetic resonance may occur at the position in which the peripheral portion of the chamber space CH and the boundary S of the sheath region meet. As a result, an etching rate of the central portion of the substrate W in FIG. **8**C may be lower than an etching rate of the central portion of the substrate W in FIG. **8**A.

[0093] Referring to FIGS. **8**A to **8**C, the second current value set A**2**, for which magnetic resonance occurs at a position at which the central portion of the chamber space CH and the boundary S of the sheath region meet, may be selected. As a result, process result values in the central portion of the substrate W may be controlled upward.

[0094] In some implementations, the control unit **190**, as described with reference to FIG. **1**, may control current values flowing in the coils **151**, **152**, and **153**, as illustrated in FIGS. **7**A to **7**C. For each of a plurality of current value sets including various combinations of currents, a resonance position, having a magnetic flux density for which the oscillation period of the electrons and the rotation period of the electrons match, may be determined for each oscillation period of the electrons (e.g., 120 MHz, 60 MHz, etc.). Then, the control unit **190** may select a current value set for which the target position matches the resonance position, among the plurality of current value sets, and may control currents corresponding to the current value set to flow in the coils **151**, **152**, and **153**, to generate magnetic resonance at the target position.

[0095] A thickness of the sheath region SH and/or a magnetic flux density for each position of the chamber space CH may change depending on variability (e.g., errors) in a process environment. Depending on the variability in the process environment, the current value set for generating magnetic resonance at the position at which the boundary S of the sheath region and the central axis of the chamber space CH meet may be changed.

[0096] According to some implementations, a plasma processing device **100** monitors a plasma process in real time, and may change the current value set for generating magnetic resonance according to monitoring results.

[0097] For example, as shown in FIG. **9**, a plasma processing device **100** includes a chamber space CH, an upper electrode **130**, a first power supply **131**, an electrostatic chuck **140**, a second power supply **141**, a substrate W, an electromagnet **150**, a third power supply **160**, an optical interface **181**, a spectrometer **182**, and a control unit **190**, as described with reference to FIG. **1**.

[0098] As described with reference to FIGS. **5** to **8**, the plasma processing device **100** may control a magnetic field formed by the electromagnet **150** to generate magnetic resonance at a target position at which a central axis of the chamber space CH and a boundary of a sheath region meet. Moreover, the magnetic field formed may be controlled. For example, the third power supply **160** may control currents a**1**, a**2**, and a**3** to have current values calculated to cause the magnetic resonance at the target position. As discussed above, the magnetic flux density can depend on at least an oscillation period of electrons, which may be determined by at least one of a first frequency F**1** and a second frequency F**2**.

[0099] In some cases, even though the calculated currents a**1**, a**2**, and a**3** flow through the coils **151**, **152**, and **153**, magnetic resonance may not occur at the target position in the chamber space CH due to a change in process environment. For example, there may be a fine difference in process environment in different chambers, and the process environment may be changed over time even in the same chamber. Depending on the process environment, a thickness of the sheath region SH or the like may differ from the calculated or assumed one, and, when the currents a**1**, a**2**, and a**3** are applied, the position at which magnetic resonance actually occurs may deviate from the target position. When the position in which magnetic resonance actually occurs is different from the target position, it may be difficult to effectively control process distribution in a central portion of the substrate W.

[0100] According to some implementations, the plasma processing device **100** perform OES monitoring using the spectrometer **182**, and adjusts values of currents applied to the coils **151**, **152**, and **153** according to the monitoring results, to effectively control process distribution.

[0101] Plasma may emit light with a unique wavelength, depending on a type of plasma gas or a type of a by-product participating in a plasma process. The optical interface **181** may include an optical window through which light in the chamber space CH may be transmitted. The optical interface **181** may monitor light generated from the plasma in the chamber space CH. Additionally, the spectrometer **182** may split the light according to a wavelength, and may then monitor an intensity of the light for each wavelength.

[0102] According to some implementations, the control unit **190** monitors an intensity of light at a target wavelength corresponding to a unique wavelength of light emitted from the by-product, and, when a measured value of the intensity of light is different from a reference value, values of the currents a**1**, a**2**, and a**3** flowing in the coils **151**, **152**, and **153** may be adjusted in a manner such that a difference $\Delta\tau$ between the measured value and the reference value decreases. The control unit **190** may control the third power supply **160** such that adjusted currents a**1'**, a**2'**, and a**3'** flow in the coils **151**, **152**, and **153**, to control to generate magnetic resonance at the target position.

[0103] FIGS. **10**A and **10**B illustrate OES monitoring results of a spectrometer **182**, as described with reference to FIG. **9**. A graph in FIG. **10**A illustrates an intensity of light by wavelength at a certain point in time. As previously explained, light having a unique wavelength may be emitted, depending on a type of by-product participating in a plasma process. The intensity of light at the unique wavelength may be changed, depending on an amount of the by-product generated in the plasma process. Therefore, when the unique wavelength is determined as a target wavelength TW and the intensity of light at the target wavelength TW is analyzed, it may be determined whether a process reaction has been actively performed.

[0104] When magnetic resonance occurs at a target position in which a central portion of a chamber space CH and a boundary of a sheath region SH meet, a large amount of plasma may be generated at the target position, and an active process reaction may occur in a central portion of a substrate W. Due to the active process reaction, a large amount of a by-product may be generated, and an intensity of light at the target wavelength TW may be relatively large. When magnetic resonance does not occur at the target position, an intensity of light at the target wavelength TW may be relatively small.

[0105] According to some implementations, a plasma processing device **100** determines the intensity of light at the target wavelength TW when magnetic resonance occurs at the target position as a reference value, and may adjust current values of coils **151**, **152**, and **153**, based on the difference value Δτ between the reference value and the measured value (observed value).

[0106] The reference value may be determined in advance. As a first example, current values determined in the same manner, as described with reference to FIGS. **5** to **8**, may be applied to coils of a plasma processing device in an experimental environment, and a reference value may be determined by analyzing an intensity of light at a target wavelength TW using an optical interface **181** and a spectrometer **182** of the plasma processing device. As a second example, a reference value may be determined by applying the current values to simulate an intensity of light measured at a target wavelength TW.

[0107] The graph in FIG. **10**B illustrates an intensity of light at the target wavelength over time. A process environment of the plasma processing device **100** may differ from an experimental environment, and the process environment may be changed, depending on time. Therefore, a difference value Δτ may occur between the reference value and the measured value.

[0108] As illustrated in FIG. **10**B, the spectrometer **182** may monitor the intensity of light at the target wavelength over time. According to some implementations, the plasma processing device **100** may monitor the intensity of light at the target wavelength in real-time while adjusting the current values of the coils **151**, **152**, and **153**, to determine current values minimizing the difference value Δτ.

[0109] FIG. **11** is a flowchart illustrating a method of controlling a magnetic field according to some implementations. The method includes S**210** to S**240**, and S**210** to S**240** may be performed by a control unit **190**, as described with reference to FIGS. **1** and **9**.

[0110] In S**210**, the control unit **190** sets initial current values of coils **151**, **152**, and **153**. For example, the control unit **190** may set current values calculated in the same manner as described with reference to FIGS. **5** to **8**, as the initial current values of the coils **151**, **152**, and **153**.

[0111] In S**220**, the control unit **190** controls a plasma processing device **100** to perform a plasma process, based on the set initial current values. As the plasma process progresses, a by-product may be generated in a chamber space CH.

[0112] In S**230**, the control unit **190** monitors OES data in real time using a spectrometer **182**. Specifically, the control unit **190** may determine a unique wavelength generated by the by-product as a target wavelength, and may monitor an intensity of light at the target wavelength over time.

[0113] In S**240**, the control unit **190** determines a difference value Δτ between a reference value and a measured value in the intensity of light in a target wavelength.

[0114] In S**250**, the control unit **190** corrects the current values of the coils **151**, **152**, and **153** such that the difference between the reference value and the measured value decreases.

[0115] FIG. **12** is a flowchart detailing a method of correcting current values according to some implementations.

[0116] An operation of correcting current values, S**250**, as described with reference to FIG. **11**, may include S**251** to S**259**. S**251** to S**259** represent operations of controlling a difference value Δτ between a reference value and a measured value, so as to minimize the difference value, while adjusting current values of coils **151**, **152**, and **153** (e.g., one by one).

[0117] In S**251**, a control unit **190** updates a difference value Δτ determined in real time.

[0118] In S**252**, the control unit **190** adjusts a first current value a**1** of a first coil **151**. For example, the control unit **190** may monitor whether the difference value Δτ decreases while increasing or decreasing the first current value a**1**, and may adjust the first current value a**1** in a manner in which the difference value Δτ decreases.

[0119] In S**253**, the control unit **190** determines whether the difference value Δτ converges to a minimum value by adjusting the first current value a**1**. When the difference value Δτ does not converge ("No" in S**253**), the control unit **190** may repeatedly perform S**252** and S**253** until the difference value Δτ converges to the minimum value. When the difference value Δτ converges to the minimum value and the difference value Δτ no longer changes in a decreasing manner ("Yes" in S**253**), the control unit **190** updates the first current value a**1** to a first current value a**1'** at which the difference value Δτ converges to the minimum value.

[0120] In S**254**, the control unit **190** updates a difference value Δτ determined in real time.

[0121] In S**255**, the control unit **190** adjusts a second current value a**2** of a second coil **152**. Similar to what was described in S**252**, the control unit **190** may monitor whether the difference value Δτ decreases while increasing or decreasing the second current value a**2**, and may adjust the second current value a**2** in a manner such that the difference value Δτ decreases.

[0122] In S**256**, the control unit **190** determines whether the difference value Δτ converges to a minimum value by adjusting the second current value a**2**. When the difference value Δτ does not converge ("No" in S**256**), the control unit **190** may repeatedly perform S**255** and S**256** until the difference value Δτ converges to the minimum value. When the difference value Δτ converges to the minimum value ("Yes" in S**256**), the control unit **190** may update the second current value a**2** to a second current value a**2'** when the difference value Δτ converges to the minimum value.

[0123] In S**257**, the control unit **190** updates a difference value Δτ determined in real time.

[0124] In S**258**, the control unit **190** adjusts a third current value a**3** of a third coil **153**. Similar to what was described in S**252**, the control unit **190** may monitor whether the difference value Δτ decreases while increasing or decreasing the third current value a**3**, and may adjust the third current value a**3** such that the difference value Δτ decreases.

[0125] In S**259**, the control unit **190** determines whether the difference value Δτ converges to a minimum value by adjusting the third current value a**3**. When the difference value Δτ does not converge ("No" in S**259**), the control unit **190** may repeatedly perform S**258** and S**259** until the difference value Δτ converges to the minimum value. When the difference value Δτ converges to the minimum value ("Yes" in S**259**), the control unit **190** may update the third current value a**3** to a third current value a**3'** when the difference value Δτ converges to the minimum value.

[0126] When the difference value Δτ converges to '0' or another target value or range while performing S**251** to S**259**, the control unit **190** may end the correction, and may update the current values of the coils **151**, **152**, and **153** to current values of the coils **151**, **152**, and **153** when the difference value $\Delta\tau$ converge to '0' or the other target value or range.

[0127] Referring to FIG. **9**, among the plurality of coils disposed in concentric circles centered on the central axis of the chamber space CH in the electromagnet **150**, the first coil **151** may be a coil disposed in the outermost concentric circle, the second coil **252** may be a coil disposed in a middle concentric circle, and the third coil **253** may be a coil disposed in the innermost concentric circle. When an amount of change in current values at the first to third coils **151** to **153** are the same, a change in magnetic flux density due to the first coil **151** may be the smallest, and a change in magnetic flux density may increase in order from the second coil **152** to the third coil **153**.

[0128] According to some implementations, in S**251** to S**259**, the control unit **190** perform current correction in a predetermined order from the outermost coil to the innermost coil. That is, the control unit **190** may select at least one of the plurality of coils in order of decreasing coil radius, and repeat the operation for the at least two of the plurality of coils. For example, the control unit **190** may preferentially perform fine adjustment of a magnetic flux density, and when magnetic resonance cannot be achieved by the fine adjustment, the magnetic flux density may be adjusted in a wider range. However, the present disclosure is not limited thereto, and the control unit **190** may perform coarse-fine tuning by performing current correction in order from the innermost coil to the outermost coil.

[0129] According to some implementations, as described with reference to FIGS. **9** to **12**, a plasma processing device **100** may generate magnetic resonance at a target position despite errors in a process environment or changes in the process environment over time, to control plasma density in a manner in which the plasma density increases in the central portion of the substrate W. As a result, a distribution control range of the plasma processing device **100** may be improved, and the improved distribution control range may be maintained despite changes in the process environment. For example, tool-to-tool matching (TTTM) of the plasma processing device may be improved, and constancy of process distribution may be improved.

[0130] FIG. **13** illustrates process parameters that result from applying current value sets having various combinations of current values to coils **151**, **152**, and **153**, as described with reference to FIGS. **5** and **9**, in a plasma processing device **100**. In the graph, the horizontal axis represents a relative position from a central axis of a substrate W, and the vertical axis represents an etching rate. On the horizontal axis, '0' represents a central portion of the substrate W, and '**-150**' and '150' represent both end portions of the substrate in any direction, parallel to an upper surface of the substrate.

[0131] A reference distribution map A**0** illustrated in a thick line in FIG. **13** represents an example distribution map when no current is applied to the coils **151**, **152**, and **153**. A lowest distribution map (AMIN) represents an example distribution map in which an etching rate in the central portion of the substrate W is minimized by applying a certain current value set to the coils **151**, **152**, and **153**.

[0132] As described with reference to FIG. **3**, when a density of plasma is controlled only by relying on Lorentz force, the density of the plasma may be controlled only in a direction of moving generated plasma from the central portion to a peripheral portion in the substrate W. Therefore, process distribution may be controlled only in a manner in which the etching rate in the central portion of the substrate W decreases as the magnetic flux density increases by applying a strong magnetic field to the chamber space CH. Comparing the reference distribution map A**0** and the lowest distribution map (AMIN), the etching rate in the central portion of the substrate W may be controlled in a range of 0 a.u. to 400 a.u. in a decreasing manner. When the etching rate in the central portion of the substrate W is lower in the reference distribution map A**0**, it may be difficult to equally control the etching rate of the central and peripheral portions of the substrate W.

[0133] According to some implementations, the plasma processing device **100** may locally increase plasma generation efficiency by causing magnetic resonance on a boundary of a sheath region, and process distribution may be controlled in a manner in which an etching rate in a local region increases. The plasma processing device **100** may control current values of the coils **151**, **152**, and **153** such that magnetic resonance occurs at a position at which the boundary of the sheath region and a central axis of the chamber space CH meet, to control process distribution in a manner such that an etching rate in the central portion of the substrate W increases.

[0134] A highest distribution map (AMAX) represents an example distribution map in which an etching rate in the central portion of the substrate W is maximized by applying a certain current value set to the coils **151**, **152**, and **153**. Comparing the reference distribution map A**0** and the maximum distribution map (AMAX), the etching rate in the central portion of the substrate W may be controlled in ranges of 0 a.u. to 140 a.u. in an increasing manner.

[0135] According to some implementations, the plasma processing device **100** may increase a control range by 35% from 400 a.u. to 540 a.u. using magnetic resonance. In addition, the plasma processing device **100** may control the etching rates of the central and peripheral portions of the substrate W to be equal or substantially equal even when the etching rate in the central portion of the substrate W is lower in the reference distribution map A**0**.

[0136] FIG. **14** illustrates a distribution of CD as a function of distance (radius) from a central axis of a substrate W. For example, a position with a radius of '0' represents a center of the substrate W, and a position with a radius of '150' represents an edge of the substrate W. The CD in FIG. **14** is provided by magnetic field control performed by a magnetic field control device in a plasma processing device having the CD, as in FIG. **4**.

[0137] According to some implementations, a control unit **190**, as described with reference to FIGS. **1** and **9**, acquires a distribution map indicating process values according to a distance from the center of the substrate W. As illustrated in FIG. **4**, when the process value is lower toward the center of the substrate W, the control unit **190** may control current values flowing coils **151**, **152**, and **153** to generate magnetic resonance at a target position at which a boundary of a sheath region and a central axis of a chamber space meet. Plasma generation efficiency in the central portion of the substrate W may be improved due to the magnetic resonance at the target position, and the CD in the central portion may be adjusted upward as an etching rate in the central portion is adjusted upward.

[0138] In some implementations, the control unit **190** may control current value flowing through the coils **151**, **152**, and **153** to avoid occurrence of the magnetic resonance at the target position, when the process value is lower as a distance from the center of the substrate W increases in the distribution map.

[0139] Comparing FIGS. **4** and **14**, in FIG. **4**, a 3 sigma value for the distribution of CD values may have a difference of 1.65 a.u. from an average value of 23.67 a.u., while in FIG. **14**, a 3 sigma value may have a difference of 1.18 from an average value of 23.83 a.u.. For example, the 3 sigma value may be reduced by 28% by adjusting the CD at the center of the substrate W upward as described herein. As a result, the CD for each position of the substrate W may be relatively uniform.

[0140] Accordingly, in some implementations, the plasma processing device **100** controls process distribution to be uniform in response to various types of process distribution maps that a chamber space may have.

[0141] FIG. **15** is a flowchart illustrating a plasma processing method according to some implementations. The plasma processing method using a plasma processing device **100**, as described with reference to FIGS. **1** to **14**, may include S**310** to S**350**.

[0142] In S**310**, a substrate W on which a material film, for example, an oxide film or a nitride film is formed, is loaded on an electrostatic chuck **140** in a chamber space CH.

[0143] In S**320**, a pressure of the chamber space CH and a temperature of the electrostatic chuck **140** are set to predetermined values. The pressure of the chamber space CH and the temperature of the electrostatic chuck **140** may be process parameters, and may be changed during plasma processing.

[0144] In S**330**, a process gas is injected into the chamber space CH. The process gas may be injected into the chamber space CH using a gas supply device **120**, as described with reference to FIG. **1**.

[0145] In S**340**, the process gas injected into the chamber space CH is converted into plasma, and the material film on the substrate W is plasma processed. The process gas may be converted into plasma by a first high-frequency power source and/or a second high-frequency power source, applied singly or simultaneously, to an upper electrode **130** and a lower electrode including the electrostatic chuck **140**, as described with reference to FIG. **1**.

[0146] According to some implementations, a magnetic field is formed in the chamber space CH by coils included in a magnetic field control device, and a process value for multiple positions of the substrate W may be controlled by the magnetic field.

[0147] For example, according to some implementations, when the process value (e.g., etch rate) is lower toward the central portion of the substrate W, a current value flowing through the coils may be controlled such that the magnetic field is generated at a target position at which a boundary of a sheath region of the chamber space CH and a central axis of the chamber space CH meet. As magnetic resonance occurs at the position, plasma density at the position may increase, and the process value in the central portion of the substrate W may be controlled upward, thereby making the process value for each position of the substrate W uniform.

[0148] In addition, or alternatively, OES monitoring of by-products resulting from plasma processing may be performed in real time during plasma processing, and the current value flowing through the coils may be adjusted according to results of the OES monitoring.

[0149] The plasma processing may include a process of etching a material film formed on the substrate W, and/or a process of forming a thin film. In some implementations, the plasma processing includes a process of chemically etching the material film formed on the substrate W, and/or a process of physically etching the material film.

[0150] In S**350**, the plasma processing may be completed by unloading plasma processed substrate W from the chamber space CH.

[0151] The plasma processing devices described herein may control a magnetic flux density of a magnetic field formed in a chamber space by controlling currents flowing in a plurality of coils at an upper portion of the chamber, and may generate magnetic resonance on a boundary of a sheath region by controlling the magnetic flux density. The magnetic resonance may be generated at a central portion of a substrate to increase plasma density in the central portion of the substrate.

[0152] The plasma processing devices described herein may control plasma density distribution not only such that plasma density in a central portion of a substrate decreases but also such that the plasma density in the central portion of the substrate increases, and may thus broadly control process distribution determined by the plasma density. Therefore, the plasma processing device may uniformly control the process distribution in response to various types of process distribution maps that a chamber space may have.

[0153] The plasma processing devices described herein may perform optical emission spectroscopy (OES) monitoring, and may correct a current flowing through the plurality of coils in real time to uniformly control process distribution, e.g., to account for differences in process environments of different chambers, process environments changing over time, and the like. Therefore, tool-to-tool matching (TTTM) may be improved, and constancy of process distribution may be improved.

[0154] While this disclosure contains many specific implementation details, these should not be construed as limitations on the scope of what may be claimed. Certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a combination can in some cases be excised from the combination, and the combination may be directed to a subcombination or variation of a subcombination.

[0155] While various examples have been illustrated and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of this disclosure.

What is claimed is:

1. A plasma processing device comprising:

a chamber body defining a chamber space;

an electrostatic chuck arranged to support a substrate in the chamber space;

an upper electrode located at an upper portion of the chamber body;

a multiple high-frequency power supply including:

a first power supply configured to apply a first high-frequency power to the upper electrode, and

a second power supply configured to apply a second high-frequency power to the electrostatic chuck,

wherein at least one of the first high-frequency power or the second high-frequency power excites a process gas supplied to the chamber space to form a bulk plasma region and a sheath region in the chamber space, and

wherein the multiple high-frequency power supply is configured to apply the at least one of the first high-frequency power or the second high-frequency power singly or simultaneously;

a magnetic field control device including at least one coil located above the upper electrode, wherein the magnetic field control device is configured to form a magnetic field in the chamber space using at least one current flowing through the at least one coil; and

a control unit configured to control the at least one current flowing through the at least one coil such that a magnetic flux density at a target position on a boundary of the sheath region has a value that results in (i) an electronic rotation period at the target position due to the magnetic field matching (ii) an electronic oscillation period that depends on the at least one of the first high-frequency power and the second high-frequency power.

2. The plasma processing device of claim **1**, wherein the target position is located where the boundary of the sheath region and a central axis of the chamber space meet.

3. The plasma processing device of claim **1**, wherein the control unit is configured to:

determine the value of the magnetic flux density that results in the electronic rotation period matching the electronic oscillation period;

determine a thickness of the sheath region;

determine at least one value of the at least one current that causes the value of the magnetic flux density to be present at the boundary of the sheath region along a vertical axis passing through a common midpoint of the at least one coil, and

flow the at least one current having the at least one value into the at least one coil.

4. The plasma processing device of claim **3**, wherein the control unit is configured to determine the thickness of the sheath region based on at least one of: a temperature of the chamber space, a potential difference between the boundary of the sheath region and the electrostatic chuck, a pressure of the chamber space, and a flow rate of the process gas.

5. The plasma processing device of claim **4**, wherein the control unit is configured to adjust a determined value of the thickness of the sheath region based on process results.

6. The plasma processing device of claim **3**, wherein the control unit is configured to determine the at least one value based on a distance between the at least one coil and the boundary of the sheath region.

7. The plasma processing device of claim **1**, wherein the control unit is configured to:

determine, for each set of at least one candidate current value of multiple sets of at least one candidate current value, a corresponding resonance position having, when the set of at least one candidate current value flows through the at least one coil, the value of the magnetic flux density that results in the electronic oscillation period matching the electronic rotation period;

determine the boundary of the sheath region, wherein determining the boundary of the sheath region comprises determining a thickness of the sheath region;

select, from among the multiple sets of at least one current value, a set for which the boundary of the sheath region matches the resonance position corresponding to the set; and

flow the at least one candidate current value set into the at least one coil.

8. The plasma processing device of claim **1**, wherein the at least one coil comprises a plurality of coils having different radii, the plurality of coils having a common midpoint that is aligned with a center position of the substrate, and

wherein the magnetic field control device comprises a plurality of power sources configured to independently control magnitudes and directions of currents flowing through the plurality of coils.

9. The plasma processing device of claim **1**, wherein the control unit is configured to determine the electronic oscillation period based on at least one of a first frequency of the first high-frequency power, a second frequency of the second high-frequency power, or a third frequency that is a common multiple of the first frequency and the second frequency.

10. The plasma processing device of claim **1**, wherein the upper electrode and the electrostatic chuck face each other.

11. A plasma processing device comprising:

a chamber body defining a chamber space;

an electrostatic chuck arranged to support a substrate in the chamber space;

an upper electrode located at an upper portion of the chamber body;

a multiple high-frequency power supply including:

a first power supply configured to apply a first high-frequency power to the upper electrode, and

a second power supply configured to apply a second high-frequency power to the electrostatic chuck,

wherein at least one of the first high-frequency power or the second high-frequency power excites a process gas supplied to the chamber space to form a bulk plasma region and a sheath region in the chamber space, and

wherein the multiple high-frequency power supply is configured to apply the at least one of the first high-frequency power or the second high-frequency power singly or simultaneously;

an optical interface configured to receive light from the chamber space;

a spectrometer configured to monitor a light intensity at a target wavelength of the light;

a magnetic field control device including at least one coil located above the upper electrode, wherein the magnetic field control device is configured to form a magnetic field in the chamber space using at least one current flowing through the at least one coil; and

a control unit configured to:

set at least one initial value of the at least one current,

cause a plasma process to be performed on the substrate, and

based on a measured value of the light intensity at the target wavelength being different from a reference value, adjust the at least one current such that a difference between the measured value and the reference value decreases.

12. The plasma processing device of claim **11**, wherein the control unit is configured to determine the at least one initial value such that a magnetic flux density on a boundary of the sheath region has a value that results in (i) an electronic rotation period at a target position due to the magnetic field matching (ii) an electronic oscillation period that depends on the at least one of the first high-frequency power or the second high-frequency power.

13. The plasma processing device of claim **11**, wherein the reference value comprises an intensity value acquired in an experimental process environment or by simulation, based on the at least one initial value.

14. The plasma processing device of claim **11**, wherein the reference value is an intensity value at which magnetic resonance occurs on a boundary of the sheath region.

15. The plasma processing device of claim **11**, wherein the target wavelength comprises a light wavelength emitted by a by-product generated by the plasma process.

16. The plasma processing device of claim **11**, wherein the at least one coil comprises a plurality of coils having different radii, wherein the plurality of coils have a common midpoint aligned with a center position of the substrate, and

wherein the control unit is configured to:

determine the difference,

select a first coil from the plurality of coils, and

repeat an operation of adjusting a current flowing in the first coil until the difference converges to a minimum value or converges to a target value or target range.

17. The plasma processing device of claim **16**, wherein the control unit is configured to select at least one of the plurality of coils in order of decreasing coil radius, and repeat the operation for the at least one of the plurality of coils.

18. A plasma processing device comprising:

a chamber body defining a chamber space;

an electrostatic chuck arranged to support a substrate in the chamber space;

an upper electrode located at an upper portion of the chamber body;

a multiple high-frequency power supply including:

a first power supply configured to apply a first high-frequency power to the upper electrode, and

a second power supply configured to apply a second high-frequency power to the electrostatic chuck,

wherein at least one of the first high-frequency power or the second high-frequency power excites a process gas supplied to the chamber space to form a bulk plasma region and a sheath region in the chamber space, and

wherein the multiple high-frequency power supply is configured to apply the at least one of the first high-frequency power or the second high-frequency power singly or simultaneously;

a magnetic field control device including at least one coil located above the upper electrode, wherein the magnetic field control device is configured to form a magnetic field in the chamber space using at least one current flowing through the at least one coil; and

a control unit configured to:

acquire a distribution map indicating a process value as a function of a distance from a center of the substrate, and

based on when the process value decreasing toward the center of the substrate in the distribution map, determining at least one value of the at least one current flowing through the at least one coil such that magnetic resonance occurs at a target position at which a boundary of the sheath region and a central axis of the chamber space meet.

19. The plasma processing device of claim **18**, wherein the control unit is configured to control the at least one current flowing through the at least one coil such that a magnetic flux density on the boundary of the sheath region has a value that results in (i) an electronic rotation period at the target position due to the magnetic field matching (ii) an electronic oscillation period that depends on the at least one of the first high-frequency power or the second high-frequency power.

20. The plasma processing device of claim **19**, wherein the control unit is configured to, based on the process value increasing toward the center of the substrate, determine the at least one value of the at least one current flowing through the at least one coil to avoid occurrence of the magnetic resonance at the target position.

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